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Review



Hydro-acoustic optimization of propellers: A review of design methods

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ABSTRACT

In recent years, the escalation of anthropogenic noise within marine environments has become a pressing ecological concern. Despite the proliferation of guidelines aimed at mitigating this issue, their adoption remains non-compulsory. The International Maritime Organization (IMO) advocates for noise reduction through the refinement of propeller design and heightened research to curtail acoustic emissions, ensuring that such measures do not compromise propulsive efficiency. This review delves into the array of strategies devised to attenuate noise pollution. It scrutinizes the methodologies employed to evaluate the hydroacoustic attributes of propellers, including experimental and computational approaches. Furthermore, it summarizes the techniques employed in the multiobjective optimization of propellers, which involves balancing the maximization of efficiency against the minimization of noise emission. The findings highlight the necessity for a comprehensive, interdisciplinary approach to the hydroacoustic optimization of marine propellers, encompassing cavitation experiments and advanced numerical simulations. In the field of propeller optimization, the traditional reliance on genetic algorithms is giving way to the adoption of neural networks and surrogate models derived from computational fluid dynamics data, particularly when addressing complex, nonlinear phenomena such as hydro-acoustics and cavitation.

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1. Introduction

The recognition of the effects of Underwater Radiated Noise (URN) has grown significantly in recent years as a result of its detrimental impact on marine life. As stated in Di Franco et al. (2020), URN levels are currently 2–10 times higher than during preindustrial times and are expected to continue to rise in the near future. Frisk (2012) examined Gross Domestic Product (GDP), shipping (UNCTAD, 2022), and noise in their study and found that they exhibited similar evolution patterns. An increase of 3.3 dB per decade was noted; this has resulted in a situation where 97.7% of the ocean surface is influenced by stressors (Halpern et al., 2015).

Nabi et al. (2018) have indicated that the increase in URN has already negatively affected up to 119 species that rely on sound as their primary means of communication. The reported impacts on these species include masking, disruption of their communication (Weilgart, 2007; Erbe et al., 2016; Nabi et al., 2018) within a 100 km radius (Halliday et al., 2017; Erbe et al., 2016), and other potential threats to their survival (Pirotta et al., 2019; Erbe et al., 2019). Therefore, because of the potential for increased harm to marine life from URN emitted by shipping, it is essential to lower anthropogenic URN levels. Given that the propulsion system is the primary source of URN transmission into the water from vessels (Xu et al., 2024b; Fragasso et al., 2024), reducing anthropogenic URN generated by shipping can be achieved through comprehensive propeller design (Jägerbrand et al., 2019; Borelli et al., 2016). This suggests examining various novel factors such as the relevant frequencies (Yan et al., 2021), the models of sound propagation and their associated uncertainties (Farcas et al., 2016; Chou et al., 2021), and the trade-offs in propulsive efficiency while reducing noise (Leaper et al., 2014) during the design phase.

With respect to the reduction of the propeller URN, it is crucial to establish the necessary regulations and threshold values, along with the available acoustic mapping data. However, multiple authors (Jägerbrand et al., 2019; Erbe et al., 2019; Di Franco et al., 2020; Vakili et al., 2020) have noted a significant lack of recorded data on mandatory regulations, threshold limits, and sound propagation data for ships. Despite the general absence of regulations enforced by public authorities, various organizations have issued several recommendations, statements, and guidelines. These public communications can be categorized into two main areas: the risk of URN generation and propagation and the technical evaluation of the noise propagation issue. The main communications published over the past two decades are detailed in Table 1.

In the earlier context, UNEP (2018), CBD (2016), and MSFD (2022) highlight the general significance of URN and its threat to marine life, emphasizing the need for collaboration among various stakeholders and authorities in shipping to establish threshold limits for sound propagation. Furthermore, HELCOM (2017, 2021), OSPAR (2014, 2020), RE-ENV-07006 (2012), and MSFD (2014) are crucial communications on sound propagation, underscoring the necessity to enhance efforts in noise mapping. Additionally, in the subsequent context, CBD (2008, 2011, 2017), along with UN-75-239 (2021) and ACCOBAMS (2004, 2013, 2019), advocate for a technical evaluation of the URN shipping issue by intensifying research efforts aimed at its reduction. In line with the need for a technical evaluation of propeller URN reduction, various class rules have been issued (BV, 2018; DNV, 2010). Finally, it is important to acknowledge the recently established IMO (2014) guidelines for reducing URN from shipping. These guidelines not only highlight the critical role of research in achieving noise reduction goals but also emphasize the importance of experimental campaigns in towing tanks and cavitation tunnels, along with numerical tools like Computational Fluid Dynamics (CFD) and Finite Element Analysis (FEA) solvers, for effective hydro-acoustic propeller design.

As discussed earlier, there is significant awareness of the detrimental effects that URN from shipping can have on marine life and ecosystems, as well as its potential increase in the coming years due to the rise in maritime trade. This awareness is in sharp contrast to the absence of mandatory regulations issued by public authorities to address this issue. Nevertheless, the IMO and the classification societies have already provided detailed technical guidelines on how to mitigate this problem during the design phase. This paper aims to review various methods, both experimental and numerical, used for the hydroacoustic design and optimization of marine propellers, with a particular focus on unconventional geometries.

2. Review methodology

For the present research work, a review of the different methods used for hydroacoustic optimization of marine propellers was performed. The literature review work was divided into three stages: planning, information filtering, and synthesis. During the first stage, the search criteria were defined. The information gathering was done through the main scientific databases (e.g. Scopus), using “Underwater Radiated Noise”, “URN”, and “Hydro-acoustics” as keywords. The filtering and classification of the information was based on the framework displayed below. As can be observed, the information was divided into 4 areas, Propeller Design, Experimental Methods, Numerical Methods, and Optimisation Design (see Fig. 1).

The contributions of each area were also divided into sections. Hence, the propeller design area was divided into three sections depending on the noise mitigation measurements used. The experimental methods area was divided into Measurement Equipment and Acoustic Characterisation. Thus, the former corresponds to those contributions that focus on the arrangement of the equipment used to measure noise (e.g. use of hydrophone arrays, placement of the hydrophones in acoustic chambers). The latter corresponds to those works that focused on the management of background noise and reverberation during experimental campaigns. The Numerical Methods area was divided into two sections depending on the viscosity modeling. The optimization area was divided into two other sections depending on whether AI was used or not to optimize geometries.

In the upcoming sections of the present research work, these interests shown in the classification and filtering of the literature review will be further discussed.

3. Propeller design

The research community’s awareness of the Underwater Radiated Noise (URN) problem is well documented through various funded research projects over the past 15 years. Table 2 provides an overview of different research projects funded by the EU since 2009, many of which delve into aspects that can inform and enhance propeller design for hydroacoustic performance.

A common conclusion extracted from the highlighted research projects is the key importance of propeller design processes and techniques in achieving an efficient and quiet shipping. Recent studies

Table 1
URN communications, guidelines and regulations developed by public and private authorities.

Author	Title	Type	Year	Geographical area	Regulatory authority	Emergence of mandatory rules
UNEP (2018)	Threats posed to marine life in the western Indian Ocean from anthropogenic ocean noise and shipping, including ship strike	Communication	2018	International	UN	No
CBD (2016)	Scientific synthesis of the impacts of underwater noise on marine and coastal biodiversity and habitat	Communication	2016	International	UN	No
MSFD (2022)	Marine Strategy Framework Directive, Thresholds for MSFD Criteria: state of play and next steps, Technical Report	Technical report	2022	EU	EU	No
HELCOM (2017)	Regional Baltic underwater noise road-map 2015–17	Communication	2017	Baltic Area	HELCOM	No
HELCOM (2021)	Regional action plan on underwater noise	Communication	2021	Baltic Area	HELCOM	No
OSPAR (2014)	OSPAR inventory of measures to mitigate the emission and environmental impact of underwater noise	Communication	2014	North-Western Atlantic	EU	No
OSPAR (2020)	OSPAR inventory of measures to mitigate the emission and environmental impact of underwater noise	Communication	2020	North-Western Atlantic	EU	No
RE-ENV-07006 (2012)	Common noise assessment methods in Europe (CNOSSOS-EU)	Communication	2012	EU	EU	No
MSFD (2014)	Monitoring Guidance for Underwater Noise in European Seas. Part II: Monitoring Guidance Specifications	Technical report	2014	EU	EU	No
CMS-CBD (CBD, 2008)	Adverse anthropogenic marine/ocean noise impacts on cetaceans and other biota	Communication	2008	International	UN	No
CMS-CBD (CBD, 2011)	Further steps to abate underwater noise pollution for the protection of cetaceans and other migratory species	Communication	2011	International	UN	No
CMS-CBD (CBD, 2017)	Adverse anthropogenic marine/ocean noise impacts on cetaceans and other migratory species	Communication	2017	International	UN	No
UN-75-239 (2021)	Resolution adopted by the general assembly on 31 December 2020	Resolution	2021	International	UN	No
ACCOBAMS (2004)	Assessment and impact of man-made noise	Resolution	2004	Black Sea, Mediterranean Sea	ACCOBAMS	No
ACCOBAMS (2013)	Addressing the impact of anthropogenic noise	Resolution	2013	Black Sea, Mediterranean Sea	ACCOBAMS	No
ACCOBAMS (2019)	Anthropogenic noise,	Resolution	2019	Black Sea, Mediterranean Sea	ACCOBAMS	No
BV (2018)	Underwater radiated noise (URN)	Class rule	2018	International	BV	No
DNV (2010)	Silent class notation	Class rule	2010	International	DNV	No
LR (2018)	Additional Design and Construction Procedure for the Determination of a Vessel's Underwater Radiated Noise	Class rule	2018	International	Lloyds Register	No
ABS (2022)	Underwater Noise and External Airborne Noise	Class rule	2022	International	ABS	No
RINA (2024)	Rules for classification of ships, Part F, Section 25: Dolphin Quiet Ship and Dolphin Transit Ship notation	Class rule	2024	International	RINA	No
IMO (2014)	Guidelines for the reduction of underwater noise from commercial shipping to address adverse impacts on marine life	Guidelines	2014	International	UN	No

(Qiongfang et al., 2024) have agreed with this and outline the importance of considering not only the hydrodynamic performance but also cavitation and low noise generation to perform an integrated design of ship propellers. In this regard, the book by Carlton (2012a) is considered one of the most authoritative sources on propulsion theory. Additionally, Kerwin and Hadler (2010) and Kinnas (2017) should also be acknowledged for their contributions to propeller analysis theory. These works focus more on propulsive efficiency and its optimization rather than on noise generation and reduction. Therefore, it is crucial to understand which parameters and strategies can improve the acoustic footprint of marine propellers without compromising efficiency. In this context, the study by Brown (1999) serves as a valuable reference. They experimented with various parameters of a bow thruster, resulting in reduced noise levels at both low and high frequencies. Despite the innovative nature of their study, they were unable to attribute the noise reduction to specific parameters, but rather to the overall new design.

The following sub-sections will analyze the physics fundamentals of noise generation and review the different parameters, devices, and design concepts that have been shown to aid in noise reduction.

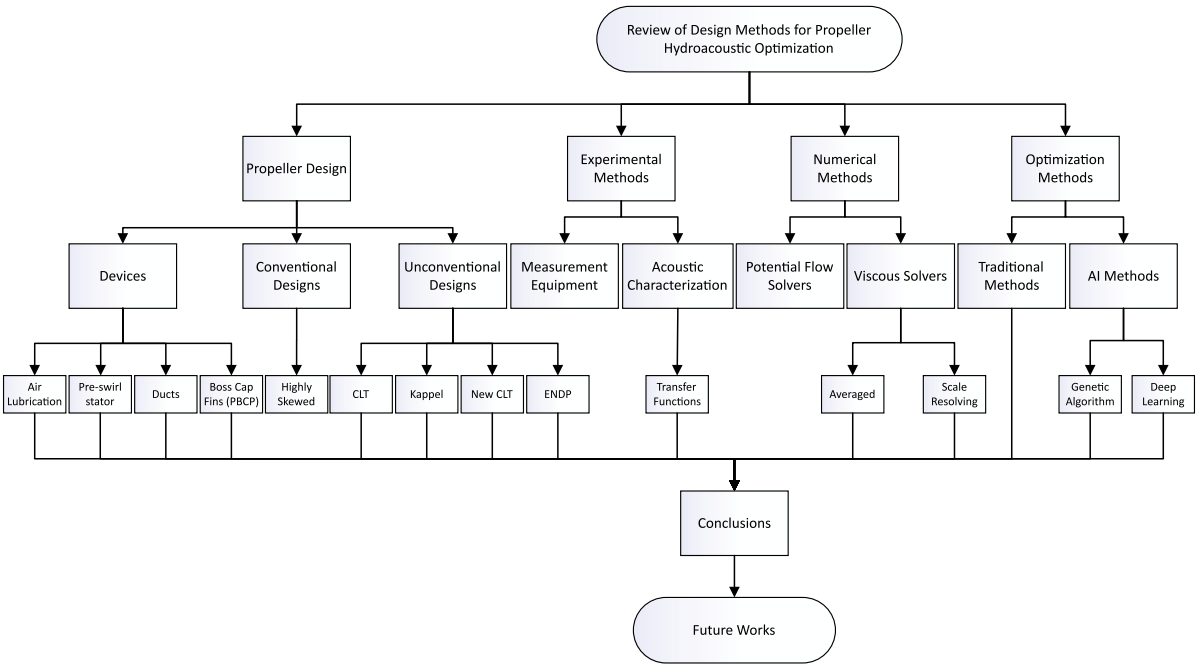


Fig. 1. Literature analysis and classification by knowledge fields, areas and tools/developed solutions.

Table 2
European research projects funded in the last 15 years.

Project name	Dates	Coordinator	Focus areas
SILENV (2009)	2009–2012	Naval Group (France)	• Ship acoustic characterization • Acoustic green label development
AQUO (2012)	2012–2015	Naval Group (France)	• Acoustic characterization • Policy markers and guidelines generation for design and operation
SONIC (2012)	2012–2015	MARIN (Netherlands)	• Ocean noise mapping • URN mitigation tools development
FIBRESHIP (2017)	2017–2020	Tecnicas y Servicios de Ingenieria SL (Spain)	• Ship design (Noise was only addressed transversely)
JONAS (2019)	2019–2022	University College of Cork UCC (Ireland)	• Noise mapping and characterization • Noise effects recognition and classification
SATURN (2021)	2021–2025	University College of Cork UCC (Ireland)	• URN mitigation tools development

3.1. Physics of underwater radiated noise in marine propellers

To understand the basics of the URN issue, it is essential to describe the core physics and mechanisms involved in the generation and propagation of underwater sound.

When examining the mechanisms of underwater sound, the initial factor to consider is the speed at which it travels. In this regard, the speed of sound in water is known to be four times that in air. It is important to note that temperature, salinity, and latitude can alter this value. To determine the exact speed of sound under specific environmental conditions, one should refer to the equation by Lovett (1978). Due to the increased propagation speed, acoustic wavelengths in water are longer than those in air, making sound transmission highly dependent on frequency. Consequently, high-frequency acoustic waves attenuate quickly, whereas low-frequency waves can travel over greater distances. Noise is typically quantified using the Sound Pressure Level (SPL), which is defined as the ratio of the acoustic pressure at a given point to a reference acoustic pressure. The reference acoustic pressure is set at the human hearing threshold (1 μPa in water and 20 μPa in air).

Concerning how noise is generated from propellers, four different main mechanisms can be identified:

- The water particles displacement due to the blade rotation motion.
- The pressure gradient between the pressure and suction side in the propeller blade.
- The cavity volume periodic fluctuations.
- The cavitation processes of bubbles and vortices originated from cavitation.

Based on this categorization of the different noise sources, propeller URN can be separated into two primary types: cavitating noise and non-cavitating noise.

Non-cavitating noise produces a noise signature for a specific propeller, which can be divided into two primary components. Firstly, discrete tonal peaks that align with the blade harmonic frequencies.

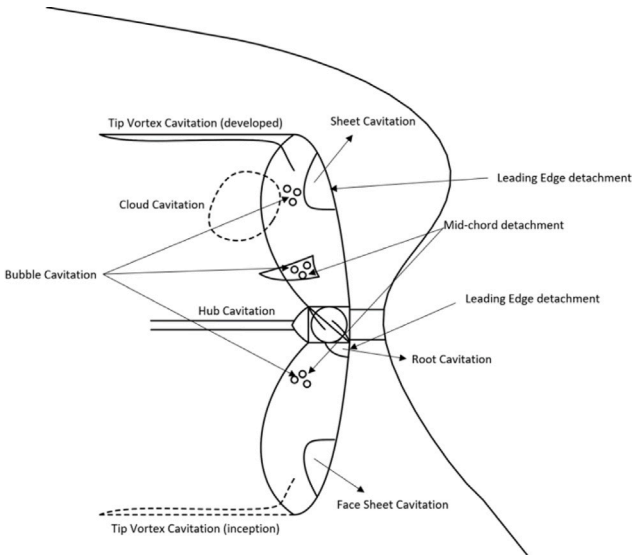


Fig. 2. Cavitation types found in marine propeller (inspired by Kinnas (2010)).

These tones have higher magnitudes at lower frequencies and occur periodically. Secondly, a broadband noise component, which can be attributed to turbulence and edge effects (such as vortex shedding and trailing edge noise).

Cavitating noise stems from cavitation phenomena, which, as stated by Carlton (2012b), occur when the fluid pressure drops below vapor pressure due to the particle acceleration caused by the propeller's rotational motion, leading to the formation of vapor bubbles, or nuclei. Cavitation inception refers to the stage where these nuclei reach a critical size and rapidly expand. The primary source of cavitating noise is the growth and collapse of these bubbles, which generate shock waves and highly unsteady pressure pulses, producing sound. Different types of cavitation exhibit distinct characteristics; therefore, when designing a propeller, it is crucial to assess which type of cavitation is most likely to occur to reduce its acoustic impact (see Fig. 2).

Steady cavitation forms, such as sheet cavitation, have been reported to cause a general increase in sound pressure levels, often perceived as white noise. In contrast, unsteady cavitation forms generate high-pressure pulses typically in the high-frequency range. This type of noise, linked to vorticity (Testa et al., 2023), is generally recognized to have weaker propagation.

Nevertheless, Li et al. (2023b) made significant discoveries regarding the transmission of cavitating sound. Their research focused on the noise generated by cavitating flow. They utilized a NACA-66 hydrofoil under both cavitating and non-cavitating conditions. It was found that sheet and cloud cavitation contribute to loading noise in the low and mid-frequency ranges. This suggests greater propagation of cavitation noise, contrary to the traditional belief that it is primarily associated with non-linear noise (quadrupole term), mainly linked with high frequencies and a rapid dissipation. This indicates an appreciable linear component associated with cavitation, suggesting more efficient propagation of certain parts of cavitating noise. Moreover, Dong et al. (2024), analyzed the acoustic-gravity wave emission from vortices generated by moving cylinders, which can be assimilated as a cavitating sound theoretical case. They found that the generated acoustic pressure concentrated in a hump below 2 kHz, thus, in the low frequency range.

3.2. Influential parameters

Based on the discussion on marine anthropogenic noise, it is clear that there is widespread agreement within the international community on the critical role of design in mitigating this issue and enhancing research efforts. Historically, propeller design research has primarily aimed at improving efficiency. On the other hand, fast merchant ships and naval vessels have also taken care to minimize acoustic emissions (Hammer and McGinn, 1978; Boswell and Cox, 1974). These vessels typically feature conventional geometries, but their blades are significantly skewed. High skew values allow the blades to move through the wakefield more gradually, thereby reducing pressure fluctuations and improving cavitation patterns.

A significant geometrical factor in propeller noise production is the flow inclination angle. An oblique inflow angle can primarily result from two sources: an inclined shaft arrangement or vessel movements under operational conditions. Several pertinent studies can be noted in this context. Bal (2020) examined the impact of the inclination angle on noise generation under cavitating conditions. To conduct their research, they used a lifting surface method, modeling the propeller blades as vortex/source element distributions. Their findings indicated a notable reduction in circulation, and consequently in propeller blade loading, with an increase in the inclination angle, particularly at the inner radii. This also led to changes in cavitation patterns, with shorter cavity lengths on the upper blades and longer cavity lengths on the lower blades. In terms of noise generation, a higher inclination angle resulted in increased overall noise levels (OASPL) for the same thrust. However, due to the numerical method used, the effect of the inclination angle on tip vortices noise could not be measured.

Aktas et al. (2016a) conducted a comprehensive experimental study on the Princess Royal twin screw vessel. Their research examined various inflow angles under atmospheric, medium, and high vacuum conditions. In all three vacuum scenarios, tip vortex cavitation was the predominant cavitation pattern observed. The study also noted a delay in the cavitation inception J value as the inclination angle increased. Similarly, Tani et al. (2018) carried out an experimental investigation about the cavitation and radiated noise patterns of a twin screw vessel's propeller under different wake conditions simulated using wire meshes. The first wake configuration represented the model scale nominal wake. The second configuration included the effects of shaft inclination and appendages, while the third configuration simulated the potential full-scale wake of the ship. The results indicated a significant impact of the inclination angle, with noise variations reaching up to 20 dB.

A recent technique used in conventional propellers to mitigate noise involves the use of different materials. Composite materials facilitate the creation of flexible blades. When the appropriate materials are selected and arranged correctly, these flexible blades can adapt their shape to loading, thus generating less noise compared to propellers made from rigid materials (Mulcahy et al., 2014). Rokvam et al. (2023) investigated the potential of composite materials for load-adaptive blades. Their study examined various bend–twist blade designs and concluded that composite propellers exhibited greater bend–twist efficiencies than those made from traditional propeller materials. Furthermore, the pitch change during periodic load variations was enhanced in composite propeller scenarios.

From this initial analysis, it is evident that there is a distinct relationship between propeller loading and the noise generated. Heavily loaded propellers generally produce more noise. Conversely, they are also associated with larger differences between the back and pressure sides of the blade, higher speeds, and consequently greater thrusts. However, these higher pressure differences can result in stronger tip vortices (Smith and Rigby, 2022). Therefore, a balance should be found between thrust requirements, or propeller loading, and the acoustic emissions of a propeller. In this context, two design strategies might help to reduce blade loading at a given speed: first, the use of energy-saving devices that reduce the required thrust for service speed, and second, varying the radial pitch distribution. The latter option may not significantly affect the generated noise, as it could involve operating in off-design conditions (Tani et al., 2015b). Based on these insights, it can be inferred that a compromise must be achieved between the energy efficiency of a propeller's geometry and its acoustic emissions.

Thus, it is crucial to develop new design methodologies and concepts that minimize the acoustic footprint of propellers without sacrificing efficiency. In this regard, unconventional geometries such as Kappel or CLT concepts, which are further discussed in the following subsection, have proven to be effective in this endeavor.

3.3. Wake improvement devices

Wake improvement devices are another method used to reduce noise in conventional propellers. These devices assist in achieving a more uniform inflow in the disk area (Carlton, 2012b). As a result, the pressure distribution becomes more uniform, and the blade pressure fluctuations can be diminished, thus reducing the risk of cavitation and cavitating noise. Among these devices, ducts (initially designed by van Lammeren (1949)) stand out as straightforward solutions to enhance the wakefield in the disk area. Ducts were originally conceived as energy-saving devices (Van Beek et al., 1986; Schneekluth and Bertram, 1998). Subsequent duct innovations, such as the Mewis Duct, have also demonstrated improvements in their acoustic emissions (Feizi Chekab et al., 2013). However, Guiard and Leonard (2013) noted that at high speeds, due to the increased risk of cavitation in the duct, the improvement in URN of a Mewis duct is more debatable. Their research also claimed that when a Mewis Duct was combined with Becker Twisted fins in a single configuration, propulsive efficiency increased by 3.5–4.2% with cavitation-free performance, which can help overcome high-speed limitations. A more recent study by Chang et al. (2019) reported that increasing the angle of attack of the inner fins in a Mewis duct can also enhance efficiency without affecting cavitation and noise.

Pre-swirl stators are another type of wake improvement device that can aid in reducing URN generation from ships. Due to the rotational motion of the blades, the wake experiences a rotational flow, leading to an energy loss (rotational loss). The preswirl stators create a counter-rotating swirling flow relative to the propeller rotation that mitigates this energy loss. Furthermore, the tip vortices were diminished or even eliminated, and TVC noise can be significantly reduced (Saettone et al., 2016). This was further explored in the numerical study conducted by Li et al. (2023a). Both broadband and tonal noise generated by a pump jet propulsor equipped with a pre-swirl stator were analyzed, and it was discovered that these wake improvement devices have more significant effects at higher speeds, particularly concerning the tonal peaks produced. Recently, Xu et al. (2024a) performed a comprehensive study of noise generation mechanisms in pre-swirl stators. In their study, they focused on pre-swirl stator pump-jet propulsors. From their study a major contribution (76%) from the stator to pressure drag was observed. In terms of the hydroacoustic analysis, the flow loss and acoustic emissions were mainly located in the rotating blades. In addition, the tip and the leading edge of the rotor blades played a dominant role in both flow losses and acoustic sources.

Propeller Boss Cap Fins (PBCF) have also demonstrated an influence on propulsor URN generation. According to Gaggero and Martinelli (2021), two distinct setups of propellers with PBCF (one with a duct and one without) were examined and compared to a standard propeller. The study found that both setups reduced noise generation in both near and far fields. However, in the near-field, the configuration without a duct exhibited superior acoustic performance (showing a 10–20 dB improvement over the ducted configuration), whereas in the far-field, the differences were less pronounced.

Recent advancements in air-lubrication aimed at reducing viscous resistance have revealed another promising method for noise mitigation. As detailed by Kawakita (2013), air lubrication systems introduce air bubbles that are distributed along the hull. These bubbles move into the propeller area on the stern's slope. This process could potentially form an air layer beneath the stern, which dampens the vibrations transmitted to the hull from cavitation and noise. Their study included performance and cavitation tests, revealing that certain void fraction conditions led to higher pressure fluctuations and an increased risk of cavitation amplification when air bubbles were injected. This risk could be mitigated by injecting the air bubbles at the center of the propeller disc and maintaining void fractions below 0.2%. Although the application of air injection systems in propellers is still under investigation, creating an air layer between the propeller and the water could potentially reduce URN.

In addition to conventional ship propulsion methods, it is crucial to consider the propulsion concept, concerning noise mechanisms. For example, Si et al. (2023) focused on ducted propellers, emphasizing the need to thoroughly analyze the vortical structures formed in ducts and their impact on acoustic emissions. Li et al. (2023c) investigated the noise generation of waterjet propulsion, discovering that, unlike conventional propellers, waterjets exhibit a significant contribution of the quadrupole noise term across all frequencies. This is highly relevant for noise propagation, as non-linear terms are typically disregarded because they become significant in propellers at high frequencies, where noise dissipates quickly.

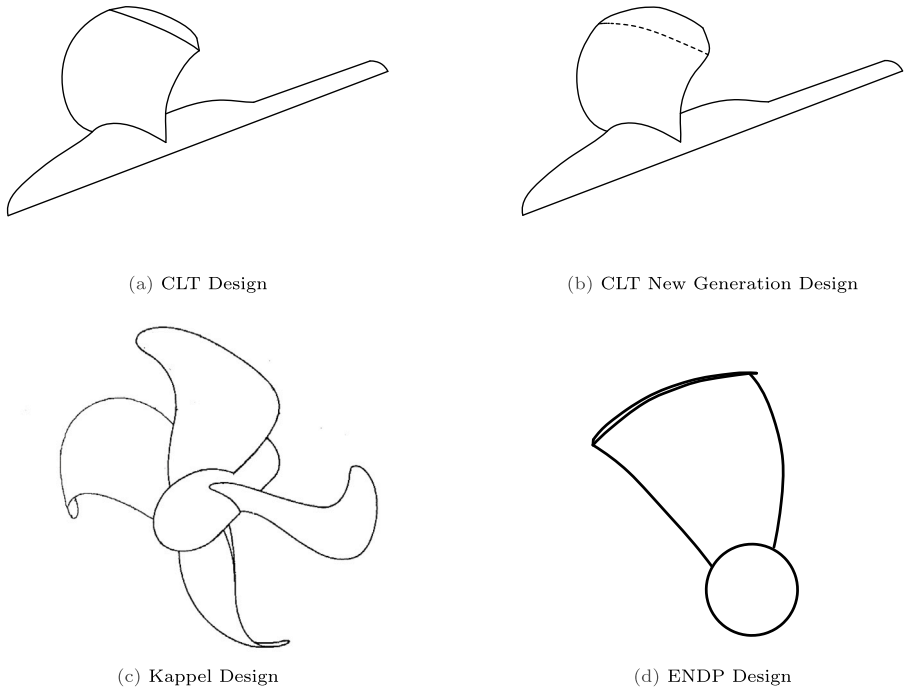
3.4. Unconventional designs

Unconventional geometries offer an alternative to mitigate noise and cavitation without compromising the propeller's energy efficiency. Various unconventional geometries have already been created to enhance efficiency and minimize noise and vibrations. One of the initial advances in these geometries was made by Perez Gomez (1976). Their research introduced a new design, the Tip Vortex Free Propeller (TVF), which employed tip bending to eliminate tip vortex cavitation.

This study was advanced by formulating a novel momentum theory (Perez Gomez and Gonzalez-Adalid, 1992) that resulted in Contracted Tip Loaded (CLT) Propellers. The new theory introduced a parameter ϵ, defined as the ratio of the suction at the front of the propeller disk to the pressure differential across the disk. Their analysis concluded that the thrust was primarily produced on the pressure side. Consequently, the under-pressure on the suction side can be easily enhanced, allowing CLT propellers to effectively manage cavitation on the suction side. Regarding Tip Vortex Cavitation (TVC), the end plate of CLT propellers creates an additional span, shifting the maximum loading towards the tip and gradually smoothing the loading curve. This theoretically reduces the intensity of vortical structures, thereby mitigating TVC.

In the initial generation of CLT propellers, it was noted that complex vortical structures emerged due to discontinuity at the junction of the blade and the tip (Gaggero et al., 2016a), leading to the formation of a secondary vortex. Gaggero et al. (2015) initially tackled the optimization of CLT propellers using genetic algorithms and subsequently developed a new generation of CLT propellers (Gaggero et al., 2016a) utilizing the same methodologies. In this updated design, the working principle of the end plate and the bending orientation remained unchanged, but the blade-tip connection was made smoother.

Kappel propellers were developed by examining the impact of their design on cavitation and noise. Initially based on Lerbs theory (Lerbs, 1952), these propellers utilized non-planar lifting surfaces instead of straight generator lines. The initial designs of the Kappel propellers (Andersen and Andersen, 1987; Andersen and Schwanecke, 1992) employed the Vortex Lattice Method to study the loading distribution and achieve appropriate blade loads. The resulting design featured a smooth bend at the tip, integrating the blades and end plates into a single unit, achieved through a new geometric parameter, the aperture angles at the blade-tip transition. These propellers demonstrated efficiency improvements of 2%–5% compared to traditional B-Series propellers. When cavitation was investigated (Andersen, 2000), they exhibited stronger tip vortex cavitation and higher cavitation-induced pressure pulses than B-Series propellers. The advancements and limitations of all these studies in Kappel propellers were summarized later in Andersen et al. (2004, 2005).



(a) CLT Design (b) CLT New Generation Design

(c) Kappel Design (d) ENDP Design

Fig. 3. Unconventional geometries (inspired by Gaggero et al. (2015, 2016a), Andersen et al. (2005) and Kehr et al. (2020)).

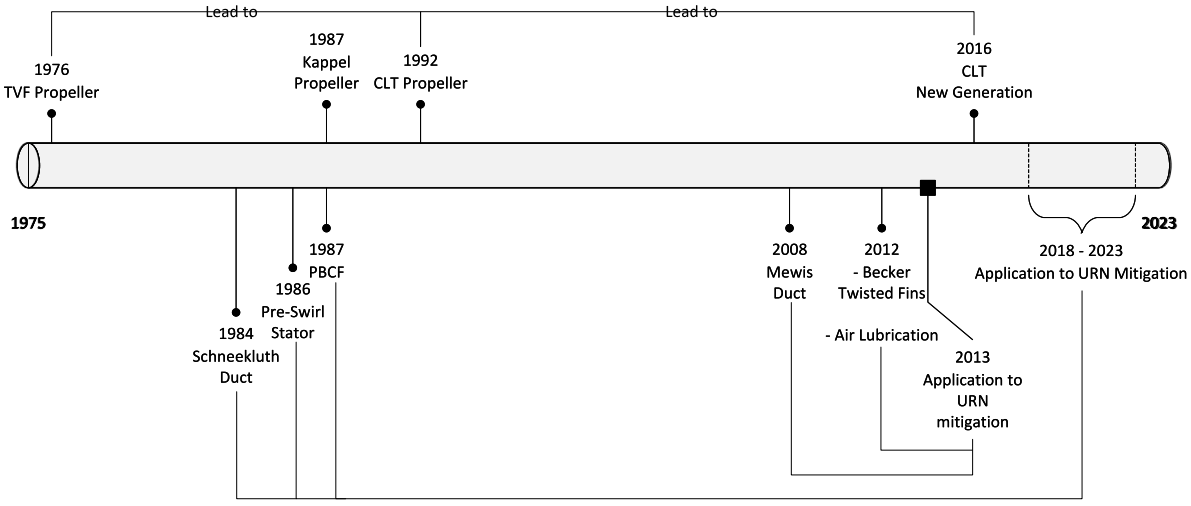


Fig. 4. Timeline of the most commonly used URN reduction solutions.

A newer concept introduced is End Plate Propellers (ENDP), featuring blades that are slightly curved towards the suction side and end plates that are bent towards the pressure side (diffused end plates) rather than having contracted tips. The testing of these propellers (Kehr et al., 2020) showed that the secondary vortex on the CLT tip plate vanished and TVC was reduced. Fig. 3 illustrates a comparison of all of the aforementioned unconventional geometries.

From the various unconventional designs created, it can be inferred that the blade tip significantly influences cavitation and noise. Inukai (2013) examined the correlation between the reduction in blade pressure fluctuations and the orientation of tip bending. In their research, the blade rake distribution was parametrically altered to create a systematic series of Backward Tip Raked Propellers (BTRP). These propellers featured tips bent towards the pressure side. This bending was intended to moderate the negative pressure on the suction side, thereby enhancing cavitation performance. Subsequently, cavitation tests were conducted on both a conventional propeller and a BTRP model. It was found that the BTRP model exhibited an efficiency increase of approximately 2.6% without significant changes in cavitation patterns compared to the conventional model. Additionally, pressure pulses decreased by almost 20% for the first blade frequency and more than 50% for the second blade frequency when comparing the BTRP with the conventional propeller. Therefore, according to Inukai (2013), since cavitation and pressure pulses are the primary sources of cavitating noise, it can be suggested that a backward tip bending orientation has the potential to improve the URN in marine propellers. A comprehensive overview of the various solutions developed to reduce URN emissions is presented in a timeline in Fig. 4.

From the reviewed literature, a clear and strong relationship emerges between pressure/vorticity fields and noise generation in marine propellers emerges. Two primary approaches are used to estimate

Table 3
Experimental research works including direct noise measurements.

Author	Infrastructure used	Size	Scale	Frequency range	Hydrophones used	Assessment of reflections
Atlar et al. (2001)	Emmerson Cavitation Tunnel	Medium 3.1 m × 1.2 m × 0.8 m	Model scale	50 Hz–100 kHz	1	Background noise measurements
Aktas et al. (2016b)	Emmerson Cavitation Tunnel	Medium 3.1 m × 1.2 m × 0.8 m	Model scale	50 Hz–20 kHz	1	Background noise measurements
Park et al. (2016)	KRISO Large Cavitation Tunnel	Large 12.5 m × 2.8 m × 1.8 m	Model scale	1–100 kHz	45 (Planar Array)	Background noise measurements
Khoo et al. (2021)	AMC Cavitation Tunnel	Medium 2.6 m × 0.6 m × 0.6 m	Model scale	1–100 kHz	1	Background noise was not reported
Park et al. (2014)	Cavitation Tunnel	Medium 1.2 m × 0.6 m × 0.6 m	Model scale	SPL results not reported	–	–
Pennings et al. (2015a)	TU-Delft Cavitation Tunnel	Small 0.6 m × 0.3 m × 0.3 m	Model scale	0–1500 Hz	1	Background noise measurements
Doolan et al. (2013)	AMC Cavitation Tunnel	Medium 2.6 m × 0.6 m × 0.6 m	Model scale	100 Hz–100 kHz	1	Background noise measurements
Park et al. (2020)	KRISO Large Cavitation Tunnel	Large 12.5 m × 2.8 m × 1.8 m	Model scale	1–80 kHz	45 (Planar Array)	Background noise estimated with transfer function corrections.
Boucheron (2020)	DGA Large Cavitation Tunnel	Large 10 m × 2 m × 1.35 m	Model scale	0–7 kHz	2	Background noise measurements not reported
Hallander (2020)	SSPA Cavitation Tunnel	Large 2.6 m width × 1.5 m height test section	Model scale	50 Hz–50 kHz	2	Background noise measurements
Tani et al. (2015a)	UniGe Cavitation Tunnel	Medium 2 m × 0.57 m × 0.57 m	Model scale	50 Hz–30 kHz	1	Background noise measurements with transfer functions
Tani et al. (2016)	UniGe Cavitation Tunnel	Medium 2 m × 0.57 m × 0.57 m	Model scale	50 Hz–30 kHz	1	Background noise measurements with transfer functions
Tani et al. (2017)	UniGe Cavitation Tunnel	Medium 2 m × 0.57 m × 0.57 m	Model scale	50 Hz–30 kHz	1	Background noise measurements with transfer functions
Tani et al. (2019)	UniGe Cavitation Tunnel	Medium 2 m × 0.57 m × 0.57 m	Model scale	50 Hz–30 kHz	1	Background noise measurements with transfer functions
Tani et al. (2020)	UniGe Cavitation Tunnel	Medium 2 m × 0.57 m × 0.57 m	Model scale	50 Hz–30 kHz	1	Background noise measurements with transfer functions
Bosschers et al. (2013)	Depressurized Model Basin MARIN	Large 248 m × 18 m × 8 m	Model scale	500 Hz–10 kHz	2	Background noise measurements
Way et al. (2021)	Depressurized Model Basin MARIN	Large 248 m × 18 m × 8 m	Model scale	1 Hz–10 kHz	3	Background noise measurements
Pang et al. (2020)	Polar Research Icebreaker Xuelong 2	Large (Measurements performed at an open field)	Full scale	1 Hz–8 kHz	2	Measurements done in open field

these fields: experimental tests focusing primarily on cavitation and numerical simulations. Recently, reduced-order models built with Artificial Intelligence (AI) have also emerged as potential tools for the prediction and optimization of the hydroacoustic performance of propellers. The following sections will analyze various experimental, numerical, and AI-based optimization studies related to the hydroacoustics of marine propellers.

4. Experimental methods

Although cavitation tunnel tests are crucial in the detailed design process of propellers, experimental campaigns that include URN measurements have only recently become more common. Prior to examining the various studies that address underwater noise in depth, a thorough overview of research works that perform direct URN measurements is presented in Table 3.

Based on this table, several preliminary observations can be made. Firstly, there is a significant agreement in covering a broad spectrum of frequencies during URN measurements. This facilitates a comprehensive characterization of marine propeller noise, as non-cavitating noise sources (e.g., blade harmonics) typically fall within the low-frequency range, whereas non-linear noise sources like cavitating noise generally occur at higher frequencies. Additionally, there is a shared understanding of the importance of evaluating background noise, given that experimental setups are often subject to certain reflections. It is noteworthy that, despite the recommendations from ITTC (2021b) regarding the use of multiple hydrophones in noise experimental campaigns, the majority of facilities employed only one hydrophone. This could be attributed to challenges in arranging them, as using an array in small and medium cavitation tunnels can be quite difficult. Finally, it is notable that there is a lack of hydro-acoustic experiments conducted in facilities other than cavitation tunnels, which could provide further insights into hull–propeller interaction in propeller noise generation. The following sub-section will analyze different studies focused on cavitation and noise prediction and reduction.

4.1. Cavitation prediction and reduction

Although direct URN measurements in marine propulsion research are relatively new, previous studies on cavitation have already highlighted the significance of noise and its connection to cavitation. Ceccio et al. (1997) investigated the influence of salt on cavitation, noting that the presence of salt and other surfactants in water alters bubble coalescence and produces foam, which directly affects the nuclei population and, consequently, cavitation. Their research found that surfactants not only altered cavitation patterns but also increased Sound Pressure Level (SPL) generation compared to freshwater.

More recently Bosschers (2015, 2019) focused primarily on cavitation in their studies, but highlighted their direct relation to noise and discussed the mechanisms of cavitating noise. During cavitation, the collapse of the cavities produces pressure pulses, which are the primary noise source. The resulting noise spectrum consists of a broadband section and tonal elements, typically found at the harmonics of the blade passage frequency. In noise reduction studies focus on noise spectrum components, the reduction of tonal noise is usually achieved by unloading the tip, thereby reducing sheet cavitation. The broadband section is primarily caused by tip vortex cavitation (TVC), which is generally the initial form of cavitation in propellers. Following this classification, some studies have focused on the reduction of TVC as an effective way to improve the acoustic emissions of propellers. The studies by Amini et al. (2019a,b) are significant, as they explored various methods (outgassing technology and winglets) to address the TVC issue. Their findings indicated that outgassing technologies merely delay TVC inception under favorable flow conditions. Conversely, the use of winglets reduced the TVC index by 68% without compromising performance. Their research did not include direct URN measurements. The works of Pennings et al. (2015b,a) are noteworthy. This Ph.D. thesis conducted an in-depth analysis of TVC as the primary source of noise and vibrations. For this research, a Small Cavitation Tunnel was utilized, incorporating stereo PIV techniques and high-speed video shadowgraphy for flow measurements, and a hydrophone for sound measurements, which was placed in a water-filled cup outside the tunnel, simulating the concept of an acoustic chamber. It is noteworthy that they showed a high potential of PIV measurements to capture the nonlinearities of the flow field under cavitating conditions. Similarly, Lee et al. (2024b) recently used visual techniques (PIV) to measure the wake flow of a submarine. Although they did not report direct URN measurements, the applicability of PIV techniques to measure unsteady flow reinforces the great potential of these tools in terms of their applicability in hydroacoustics.

Nevertheless, various researchers (e.g. Foeth et al. (2008), Huang et al. (2013) and Venning et al. (2022)) have directly investigated sheet and cloud cavitation, noting that these are more intense phenomena following TVC and can transform into shock waves, causing peaks in noise spectra. Foeth et al. (2008) examined the collapse of the sheet cavity on a 3-D foil in a Small Cavitation Tunnel, identifying sheet shedding cavitation as a significant source of noise and vibration. Subsequently, Huang et al. (2013) addressed the issue using both experimental and numerical methods to provide validation data. Venning et al. (2022) explored the impact of nuclei population on cloud cavitation in a Medium Cavitation Tunnel, utilizing high-speed cameras and nuclei injectors to test various populations, and concluded that there is a direct correlation between nuclei population and shedding, leading to shock wave generation.

From these studies two ideas can be drawn. First, it is key from the design perspective to analyze which cavitation forms are most likely to happen in a propeller as they are strongly linked with the noise spectrum it will develop. Secondly, visual techniques have shown a great potential to estimate non-linearities in the flow field of a propeller which directly impacts in the noise generation.

4.2. URN prediction and reduction

Atlar et al. (2001) conducted an experimental study primarily focused on estimating experimental noise within a Medium Cavitation Tunnel. Wake screens were employed to replicate target wake values. A notable feature of their noise measurement methodology was the use of a single hydrophone positioned inside a steel cylinder above the propeller, similar to an acoustic chamber. The data were post-processed using a filter amplifier system and ITTC corrections (ITTC, 1987), resulting in precise solutions when compared to full-scale data, despite the restricted dimensions of the cavitation tunnel ($1219 \times 806 \times 3100$ mm). Aktas et al. (2016b) indicated that while Large Cavitation Tunnels (LCT) are more equipped to deliver precise URN estimates, medium cavitation tunnels can also yield valuable data. In their investigation, dummy hulls were employed instead of wire meshes to simulate target wakes; however, for noise measurements, only a single hydrophone was utilized, positioned in a streamlined strut. In contrast, hydrophone arrays were used for full-scale measurements. A key finding from comparing model and full-scale outcomes was the significance of reverberation in experimental setups and, consequently, the assessment of background noise.

Park et al. (2016) employed a planar array of 45 hydrophones for model-scale experimental URN measurements. This research highlighted the significance of hydrophone localization in ensuring the accuracy of URN measurements. Khoo et al. (2021) underscores the necessity of studying cavitation and noise together due to their interrelation. Upon cavitation inception and nuclei activation, both cavity kinematics and acoustics are influenced by local pressure. Their study utilized high-speed cameras for cavity measurements and a single hydrophone for URN measurement, testing various nuclei population conditions. The findings indicated that both cavitation number and nuclei population directly affect SPL and noise spectra. Focusing on the importance of cavitating noise, Park et al. (2014) conducted research on scaling procedures for TVC noise in relation to the Reynolds number. While additional studies were necessary, it served as a valuable foundation for subsequent works. Bosschers (2021) also investigated the impact of the Reynolds number on TVC scaling. In their research, the Vortex Model from Bosschers (2018, 2019) was utilized to establish a connection between TVC noise scaling and speed. This scaling procedure for TVC noise was implemented on the Princess Royal Catamaran. Predictions of TVC noise using the developed scaling procedure were compared with full-scale data, showing good agreement at low-medium speeds (7 knots). However, at higher speeds (9 knots), further investigation was required.

The research by Dang et al. (2019) is noteworthy, as they explored the potential of biomimetic surfaces for noise reduction. Their study involved applying micro-grooved surfaces to a 3-D hydrofoil, which significantly reduced tonal peaks compared to the original smooth-surfaced foil. Similarly, Huang et al. (2019) investigated the impact of a T-shaped tip on broadband noise and found that it reduced dipole noise, resulting in a strong agreement between experimental and numerical results.

The issue of background noise in hydro-acoustic experiments was highlighted by Briancon et al. (2013), who reviewed various experimental noise measurement techniques. When studying the effects of reverberation and background noise, the primary focus of research over the past decade has been on developing methods to acoustically characterize different facilities. An excellent example of acoustic characterization in cavitation tunnels is provided by Doolan et al. (2013). In their study, noise was measured using a single hydrophone placed in an acoustic chamber. The significance of material selection for the acoustic chamber design was emphasized, noting that the acoustic impedance of the chamber material should match that of the fluid. Regarding the method for characterizing experimental facilities, the most common technique involves developing a transfer function for the facility. This approach assumes that the cavitation tunnel operates as a single-input

single-output system. Additionally, the assumptions of linearity and time-invariance in the system response are applied, as these are realistic for acoustic propagation. As a result, the system's output can be easily modeled.

Over the past ten years, numerous studies have examined this approach to separate reverberation and background noise from URN experimental data. Park et al. (2020) confirmed the effectiveness of a transfer function created through virtual source URN tests. Their research demonstrated that URN tests conducted with a hydrophone array, combined with extrapolation techniques, can yield precise URN measurement results. In earlier research, Park et al. (2016) employed a planar array in a large cavitation tunnel to measure URN using models of various ship types. The primary goal of their study was to develop a noise localization technique, and they also found that a hydrophone array can effectively mitigate background noise issues.

Conversely, hydrophone arrays are seldom employed, and experimental research is typically conducted with a limited number of hydrophones. For instance, the study by Pang et al. (2020) utilized two hydrophones to measure URN. They concentrated on propagation loss in shallow water and found that uncertainties were greater in the low-frequency range. Consistent with the findings of Park et al. (2020) and Boucheron (2020), the relationship between acoustic measurements and acoustic response was examined, concluding that the positions of the acoustic source and noise source are crucial in defining the transfer function. A notable study on the repeatability and sensitivity of URN experiments was performed by Hallander (2020). It is noteworthy that this study employed three hydrophones to measure URN, demonstrating that transfer functions can enhance predictions in the 400 Hz–5 kHz frequency range. Tani et al. (2015a) began addressing the impact of confined environments on URN estimation. Their research tested periodic, deterministic, and wideband signals, concluding that sine-sweep signals improved the signal-to-noise ratio for transfer-function development applications.

Subsequently, Tani et al. (2016) compared URN measurements from two different facilities (SSPA[1] and UniGe[2]) with full-scale results. This study noted high background noise levels at both facilities, yet the results still aligned well with the full-scale data. One of the final observations was that while transfer functions showed significant improvements in URN predictions under design conditions, their benefits were more debatable under off-design conditions. Later, Tani et al. (2017) conducted similar studies with comparable experimental setups and outcomes, but at different facilities (ECT[3] and UniGe). More recently, Tani et al. (2020) published a study comparing different experimental facilities in a Robin Round Test programme. In 2019, Tani et al. (2019) examined various aspects of transfer function measurements, recommending that hydrophone mounting should be performed in acoustic chambers to reduce near-field effects. They also suggested using 1/3-Octave Band Spectra to reduce uncertainties in the frequency response of the tunnel to cavitation noise. A remarkable recent study is the one performed by Lee et al. (2024a). They remarked the importance of analyzing in detail the dynamics of tip-vortex cavitation and noise scaling when estimating full-scale URN. In their study, the relationship between the size of the tip vortex cavity size and the cavitation number (which is key to scaling TVC noise) was derived from the Proctor vortex model. When comparing the predicted results with the full-scale measurements, it showed deviations below 5 dB.

While URN experiments are typically conducted in cavitation tunnels, some researchers have also measured noise in towing tanks. Bosschers et al. (2013) performed URN measurements in MARIN's depressurized wave basin, identifying wall noise reflection as a primary issue that can influence background noise and the basin's noise spectra. Their study found that above 1 kHz, these effects were minimal, but below 1 kHz, background noise was significant. Subsequently, Way et al. (2021) conducted an acoustic characterization of a towing tank using an Image Source Method. This method assumed smooth tank walls, a rectangular towing tank section, and no diffraction. The developed method showed that the Image Source method provided a good qualitative match with noise measurements, making it useful for designing hydrophone positions before experimental campaigns.

Besides the various studies conducted on cavitation and noise, it is crucial to highlight the distinct focus areas and recommendations provided by the ITTC for studying URN. In the latest report from the Specialist Committee (ITTC, 2021a), the significance of hydrophone sensitivity and calibration was stressed, recommending an estimate of sensitivity in a free field environment and at a distance of 1 m from the free surface. Their report also underscores the importance of the number and positioning of hydrophones. In the ITTC Recommended Procedures and Guidelines for Model-Scale Propeller Cavitation Noise Measurements (ITTC, 2017), the emphasis on hydrophone calibration is reiterated. The recommendation to use multiple hydrophones instead of just one in the propeller plane is also noted. Consistent with other committees, it was suggested to account for the cavitation tunnel blockage factor and to use wire screens to simulate the wake when dimensions are restricted.

Considering the aforementioned studies, several observations can be emphasized for the advancement of future research. The existing studies reviewed have primarily concentrated on evaluating the acoustic properties of cavitation tunnels, with a particular focus on the hydrophone setup and the measurement of background noise and spurious reflections, thereby neglecting some important aspects for hydro-acoustic experimental campaigns. In this context, there is a notable lack of research on noise source localization and experimental campaigns under off-design conditions. Additionally, it has been noted that the acoustic characterization of other experimental facilities, such as towing tanks, where the impact of hull–propeller interaction on URN can be further investigated, is quite innovative. Previous authors have attributed the scarcity of hydro-acoustic studies in towing tanks to the challenges posed by reverberation in these facilities.

5. Numerical methods

Numerical tools can enhance the acoustic performance of marine propellers, reducing costs and time. However, integrating acoustic analysis with CFD solvers can be computationally intensive. Therefore, it is essential to evaluate the different numerical methods available. Table 4 summarizes the primary numerical studies on propellers, encompassing various modeling and optimization techniques.

The reviewed literature revealed a notable finding. Regarding the frequency range in various studies, choosing a low minimum frequency has a negligible effect on the CPU cost of simulations. However, variations in the upper frequency limit were observed based on the numerical method used. Consequently, the studies were categorized by their numerical schemes, and their maximum frequencies were examined (see Fig. 5).

Several noteworthy aspects can be highlighted before delving deeper into the various reviewed studies. Firstly, when modeling acoustics, there is a clear preference of implementing (Ffowcs Williams and Hawkings, 1969) (FW–H) acoustic analogy coupled with the Navier–Stokes equations (flow solver) over using low fidelity or semi-empirical models. To address this coupling, the continuity and momentum equations are rearranged in wave-equation form. This equation computes the sound pressure at a certain point and divides the noise into three components:

[1] SSPA Maritime Centre Cavitation Tunnel, P.O. Box 24001, SE-400 22 Göteborg, Sweden.

[2] University of Genoa, Department of Electrical, Electronic, Telecommunication Engineering and Naval Architecture (DITEN), Naval Architecture and Marine Engineering Group.

[3] Emmerson Cavitation Tunnel, School of Engineering, Newcastle University.

Table 4
Numerical studies performed.

Author	Numerical model	Noise modeling	Cavitation	Optimization method	Geometry	Scale	Max. Frequency	Time cost
Brizzolara and Gaggero (2009)	Panel method (BEM)	–	Max. cavity area and volume were analyzed. Reductions of up to 73.4% between original and optimized	ModeFRONTIER. Genetic algorithms	CBI50 propeller with NACA a08 profile sections	Model scale	1000 Hz	Low (10–20 min)
Ebrahimi et al. (2019)	Panel method (BEM)	FW–H acoustic analogy. Quadrupole component is neglected. Reductions of 5.44 dB in SPL were observed between original and optimized geometries.	–	Genetic algorithms	DTMB 4119 propeller	Model scale	50 kHz	Low (not reported)
Lloyd et al. (2015b)	URANS ($\kappa-\omega$ SST)	FW–H acoustic analogy. Porous formulation	–	–	INSEAN E779A	Model scale	Analysis performed over time	Medium (not reported)
Shin and Andersen (2015)	DES ($\kappa-\omega$ SST)	–	Rayleigh Plesset model. Phase averaged iso-contours were extracted	–	Kappel propeller with modified tip configurations (3 configurations were analyzed)	Model scale	–	High (not reported)
Hu et al. (2021a,b)	DES ($\kappa-\omega$ SST)	FW–H acoustic analogy. Quadrupole component is neglected	Schnerr Sauer model. Cavitation is analyzed through cavity isosurfaces	–	E779A propeller	Model scale	10 kHz	High (not reported)
Sezen and Atlar (2023a,b,c)	DDES ($\kappa-\omega$ SST)	FW–H acoustic analogy. Porous formulation	Schnerr Sauer model. Phase averaged iso-contours were extracted	–	Princess Royal	Model and Full scale	10 kHz	High (not reported)
Balaras et al. (2015)	LES	–	Phase averaged and vorticity iso-contours were extracted	–	INSEAN E1619	Model scale	–	Very high (not reported)
Bensow and Liefvendahl (2016)	LES	FW–H acoustic analogy. Porous formulation.	Schnerr Sauer model	–	Princess Royal	Model scale	10 kHz	Very high (not reported)
Kimmerl et al. (2021)	I-LES. Implicit subgrid-scale modeling.	FW–H acoustic analogy. Porous formulation.	Schnerr Sauer model	–	P1595	Model scale	50 kHz	Very high (not reported)
Posa et al. (2022a,b,c,d)	LES-IB. WALE subgrid model	FW–H acoustic analogy.	Phase averaged and vorticity iso-contours were extracted	–	INSEAN E1658	Model scale	100 kHz	Very high (1.5 million hours at 1024 cores)
Posa (2022)	LES. WALE subgrid mode	–	Phase averaged iso-contours were extracted	–	INSEAN E1658 (with and without winglets)	Model scale	–	Very high (16 million hours at 2048 cores)
Dlamini et al. (2023)	LES. Smagorinsky subgrid model	FW–H acoustic analogy. Both impermeable and porous formulation	Rayleigh Plesset. Kubota derived model. Phase averaged iso-contours were extracted	–	Fixed screw propeller model. Not specified	Model scale	80 kHz	Very high (not reported)
Lafeber et al. (2022)	Panel method (BEM)	Low fidelity model	Semi-empirical model	Iterative processes	Wageningen C and D series	Model and full scale	100 kHz	Low (not reported)

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Table 4 (*continued*).

Doijode et al. (2022a,b)	Panel method (BEM)	–	–	Dynamic optimization with supervised Naive-Bayes classifiers. Halton sampling for dataset generation	B4–70	Model scale	–	Low (not reported)
Jiang et al. (2020)	Strip method	–	–	Data driven models. Neural networks and surrogate models	Not specified	Model scale	–	Low (not reported)
Miglianti et al. (2019, 2020)	Panel method (BEM)	Noise data from experiments	–	Data driven models. Artificial neural networks	–	Model scale	10 kHz	Low (not reported)
Oneto et al. (2020)	Panel method (BEM)	Noise data from experiments	–	Data driven models. Artificial neural networks	–	Model scale	10 kHz	Low (not reported)
Seol et al. (2002) y Seol et al. (2005)	Panel method (BEM)	FW–H	–	–	–	Model scale	1 kHz	Low (not reported)
Ekinci et al. (2010)	Panel method (BEM)	Low fidelity model	Cavitation patterns were estimated	–	DTMB-4119	Model scale	1 kHz	Low (not reported)
Sakamoto and Kamiirisa (2018)	DES	Low fidelity model	Cavitation patterns were estimated	–	4 different geometries	Model scale	1 kHz	High (not reported)
Belibassakis and Politis (2019)	Vortex Lattice	FW–H	Cavitation patterns were estimated	–		Model scale	100 Hz	Low (not reported)
Kim and Kinnas (2022)	BEM	FW–H	–	–	–	Model scale	–	Low (not reported)
Brizzolara et al. (2008) and Gaggero et al. (2010)	BEM/RANS	–	–	–	DTMB-4679	Model scale	–	Low/Medium (30 min/5 h 12 processors)
Ianniello et al. (2013, 2014b,a)	RANS	FW–H	–	–	INSEAN E779	Model scale	Reported over time	Medium (not reported)
Bertetta et al. (2012b)	BEM	–	Cavitation patterns were estimated	Genetic algorithms	CPP propeller	Model scale	100 Hz-	Low (not reported)
Bertetta et al. (2012a)	BEM/RANS	–	Cavitation patterns were estimated	–	CLT	Model scale	–	Low/Medium (20 h 12 processors for RANS)
Ghassemi et al. (2018)	RANS	FW–H	Cavitation patterns were estimated	–	–	Model scale	1000 Hz	Medium (not reported)
Lidtke et al. (2016a)	LES Smagorinsky subgrid model	FW–H	Cavitation patterns were estimated	–		Model scale	100 Hz	Very high (not reported)
Lidtke et al. (2022)	I-DDES	FW–H	Cavitation patterns were estimated	–	–	Model scale	10 kHz	High (not reported)
Gaggero and Ferrando (2023)	I-DDES	–	–	–	–	Model scale	–	High (not reported)
Chen et al. (2023)	Vortex Lattice	FW–H	Cavitation patterns were estimated	–		Model scale	10 kHz	Low (not reported)

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Table 4 (*continued*).

Zhang et al. (2010)	LES Smagorinsky subgrid model	FW–H	Cavitation patterns were estimated	–	–	Model scale	10 kHz	Very high (not reported)
He et al. (2020)	LES WALE subgrid model	FW–H	Cavitation patterns were estimated	–	–	Model scale	5 kHz	Very high (not reported)

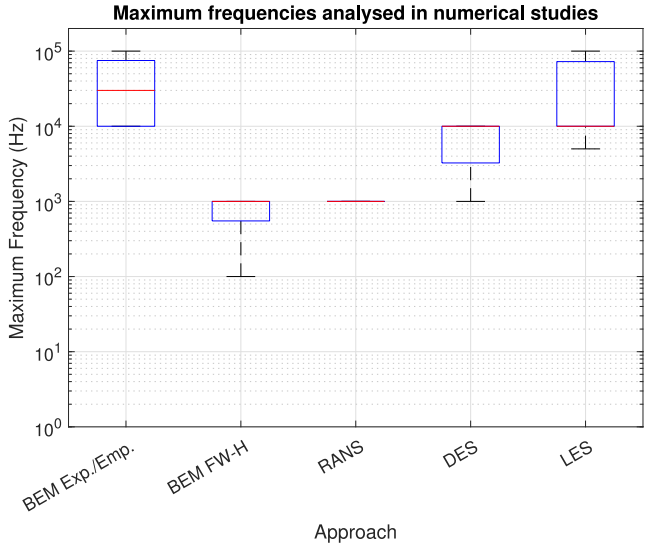


Fig. 5. Maximum frequencies analyzed in numerical studies. The median value is represented with the red line. The boxes represent the interquartile range and the whiskers represent the maximum or minimum values when they are found out of the quartiles.

- **Loading surface term**: which originates from the transient variation of forces on the propeller surface.
- **Thickness surface term**: which derives from the noise generated by the fluid displacement caused by the propeller rotation and translation motion.
- **Quadrupole volume term**: which would be originated from turbulence and non-linearities (e.g. shock waves) originating in the propeller region.

Using this equation, one can derive an acoustic pressure signal (Pa). By applying a Fast Fourier Transform (FFT) to the recorded sound pressure data, the Sound Pressure Levels (SPL) in decibels (dB) are calculated. Recent studies (Barandagh and Ghazanfarian, 2024) reinforce this preference and show the FW–H/CFD coupling as an standard framework for hydroacoustics

Secondly, there is a noticeable difference in the frequency range used when coupling BEM solvers with experimental data and/or semi-empirical models compared to acoustic analogies. In this regard, studies utilizing acoustic analogies are confined to the low-frequency range. This is because semi-empirical models and experimental data heavily rely on statistical methods, allowing for higher frequency limits to be achieved. The FW–H analogy relies on the physical principles of the problem (flow-field solution), resulting in generally lower uncertainty bounds, particularly when dealing with unconventional designs that have limited data. The limited number of studies employing a steady RANS approach is also notable. This is probably due to RANS offering a steady averaged method, which does not effectively capture pressure variations in the flow-field (especially in the high-frequency range, >1 kHz), necessitating the use of semi-empirical models. This also clarifies why in Fig. 5, RANS studies show no variation within the maximum frequency of interest, maintaining a common 1 kHz median value across the different studies analyzed. Additionally, RANS schemes have computational costs that are typically too high for systematic studies (unlike BEM), and they exhibit greater comparison deviations than scale-resolving simulations, especially under non-linear conditions like cavitating flows. Finally, a distinction is observed between DES and LES frameworks. Although both have similar median values for maximum frequencies, DES is preferred for studying lower maximum frequencies. Conversely, for wider frequency ranges, only LES schemes have been used among viscous solvers. With this in mind, it is essential to thoroughly analyze the different numerical frameworks available and the studies conducted using each of them.

5.1. Potential methods

Potential flow methods are cost-effective because they do not account for viscous stresses. The Lifting Line Method (LLM) was one of the first potential flow methods developed, followed by its derivative, the Vortex Lattice Method (VLM), which was based on LLM. Initially, these methods were applied to aerodynamics of wings (SP-405 NASA, 1976). Due to their excellent performance at low Mach numbers, they were later adopted for marine applications, as noted by Yuasa (1984). LLM models the propeller blades as a single vortex line along the span. In contrast, VLM treats the blade as a thin surface with superimposed circulation distributions. These methods were used mainly to evaluate propeller performance and did not include models to predict propeller noise.

Another method that has seen extensive use in marine propeller analysis is the Boundary Element Method (BEM). This technique normally discretizes the boundary of the case study into quadrilateral panels rather than vortices or sources such as LLM and VLM, and it solves the potential flow on that surface. Initial studies conducting hydroacoustic analysis of propellers with BEM flow solvers began by focusing solely on non-cavitating noise. Seol et al. (2002) combined a BEM solver with FW–H analogy under both uniform and non-uniform inflow conditions to predict non-cavitating noise. Consequently, the non-linear quadrupole term, primarily due to cavitating noise, was omitted. Their research indicated that the loading term was a major contributor in uniform flow conditions, whereas in disturbed flow conditions, the thickness term was more influential in the acoustic field. Although they were unable to validate their results experimentally, similar trends were observed in a subsequent study (Seol et al., 2005). However, in current hydroacoustic analyses, it is crucial to recognize that the URN of propellers is predominantly caused by cavitation. Ekinci et al. (2010) employed a BEM approach that also considers the formation of sheet cavitation. They integrated the flow solver with Brown's semi-empirical equation for the acoustic field and tested their numerical setup on various designs. All the designs examined showed OASPL predictions similar to those obtained in previous studies (Fraser, 1986), providing a conservative noise estimate. In a later study, Soydan and Bal (2019) confirmed the usefulness of Brown's formula to predict noise in the initial design stages. Sakamoto and Kamiirisa (2018) also emphasized the ability of Brown's formula to offer a conservative prediction of URN, but coupled with viscous CFD solvers (U-RANS).

Moreover, other researchers have chosen to use BEM in combination with the FW–H acoustic analogy, due to the challenges of accurately estimating URN at high frequencies using Brown's formula. Brizzolara and Gaggero (2009) utilized the BEM solver modeFRONTIER along

with a genetic optimization algorithm to create silent propellers with unconventional profile geometries (CBI50 propeller). Although they assessed noise through cavitation development and did not quantify SPL, their findings indicated a 45% and 59% reduction in the surface and volume cavity extension of the optimized propeller compared to the original, along with a 3.22% increase in efficiency. In their research, they compared the evolution of K_T and K_Q with σ against experimental results, demonstrating an overall good match, but with significantly higher deviations for $\sigma < 2$. They also compared cavitation patterns, showing a good correlation with the experiments. Regarding the computational cost of their framework, a simulation time of 10-20 min per case was reported. However, a detailed assessment of the numerical uncertainties was not provided in their study. Subsequently, Gaggero et al. (2015) employed a similar method to optimize a tip-loaded propeller design and to develop a new CLT propeller (Gaggero et al., 2016a), which showed a 70% reduction in cavity area and a 4% increase in propeller efficiency.

Subsequently, Belibassakis and Politis (2019) investigated the generation and propagation of noise in marine propellers by integrating the BEM method with the Ffowcs Williams and Hawkings (FW–H) acoustic analogy (utilizing the Farassat porous formulation). In this approach, the quadrupole non-linear terms were considered negligible. Their study included Lloyd's mirror effect and directivity measurements. This research was confined to very low frequencies, below 100 Hz, where the non-linearities of cavitation have minimal impact, yet a 20 dB/Hz increase was observed when comparing cavitating and non-cavitating conditions. Regarding the numerical framework developed, a comprehensive assessment of numerical uncertainties was not provided. Furthermore, it is important to note that the study did not thoroughly validate the numerical results for the acoustic field. A similar methodology was employed in Ebrahimi et al. (2019), but in this instance, the quadrupole non-linear terms were ignored, and the NSGA-II genetic algorithm was used to optimize their geometry. The frequency range of interest in this study spans 10 Hz–50 kHz. Compared to the experiments, they reported deviations of 4% and 7% for K_T and K_Q in the flow field solution, and a maximum deviation of 20 dB for the predicted SPL, with mean deviations of 10 dB across the entire spectrum, which is considered a reasonable agreement for hydroacoustics. However, they noted that the highest deviations occurred in the low-frequency range (50–250 Hz), where fewer non-linearities are expected. As in studies using similar frameworks, a detailed assessment of numerical uncertainties was not carried out, and the time consumption rates were not reported.

The study by Kim and Kinnas (2022) is noteworthy as it compared URN predictions in the near field under both uniform and non-uniform inflow for non-cavitating conditions using a BEM solver coupled with FW–H analogy. Their research showed good agreement for the flow field output parameters and highlighted the impact of disturbed flows on URN. While the study provided valuable insights into sound propagation and URN in off-design conditions, further research is necessary, particularly in analyzing the URN frequency spectrum. Additionally, as with other studies, a detailed assessment of numerical uncertainties and time consumption rates is required to evaluate the feasibility of these frameworks in propeller design loops. Moreover, further investigation is needed to understand the impact of hull–propeller–rudder interaction on far-field noise. Among their recommendations, they emphasized the need for deeper exploration of propeller URN in off-design and disturbed flow conditions.

The study by Brizzolara et al. (2008) is noteworthy as it compared the predictive capabilities of viscous CFD (RANS) and BEM for propeller performance. In terms of prediction accuracy, the maximum discrepancies between BEM and RANS were 0.57% for K_Q and 0.38% for K_T, but the differences in computational time were significantly greater. The RANS solver required 5 h with 12 processors to reach converged solutions, whereas the BEM solver needed only 40 min with a single processor. They examined and compared 5 DTMB propeller models, demonstrating good predictive capabilities of both methods for K_T and K_Q, as well as similar blade pressure distributions across all cases. They noted that under off-design conditions, the BEM method required certain viscous corrections, such as wake-alignment correction and coupling with a viscous boundary layer solver.

In a subsequent study (Gaggero et al., 2010), improved performance of viscous CFD was observed for highly skewed propellers, attributed to increased unsteadiness in the flow-field problem (resulting from higher blade loading and inclination angles). It should be noted that these studies also included a convergence analysis, evaluating three different grids. For a more comprehensive mesh sensitivity analysis (Least Square Root verification approach), at least four grid scenarios would be necessary. These studies did not include results concerning the prediction capability and time costs of both frameworks when considering acoustics and/or cavitation analyses. Similar findings were reported by Sezen and Bal (2020) for a DTMB-4119 propeller when comparing RANS-FWH and BEM-FWH predictions. The work by Regener (2016) is also noteworthy, as it presents a hybrid approach combining BEM and viscous CFD to investigate hull–propeller interactions and their effects on cavitation. In their research, a Reynolds Averaged Navier Stokes (RANS) modeling approach was used for the hull flow, while BEM modeling was applied to the inflow around the propeller, and the results showed good agreement with experimental data.

The work presented in Ianniello et al. (2013, 2014b,a) should be reviewed. These studies highlight the potential of BEM methods for URN investigations, noting that cavitation affects far-field noise only in the presence of shock waves. Using a viscous RANS/LES (Large Eddy Simulation) solver combined with FW–H analogy, the researchers analyzed a twin-screw ferry equipped with two CLT propellers. To account for the quadrupole noise term, a porous formulation was applied in the acoustic analysis. In their research, the numerical uncertainty of this framework was assessed, revealing a 5% bounded numerical uncertainty for the velocity field, considered acceptable. In terms of comparison with experimental data, K_T and K_Q showed good agreement, although the URN predictions were not experimentally validated. Additionally, the numerical uncertainties for the acoustic pressure field solution were not reported. It is also important to note that the framework employed, consisting of a viscous CFD solver coupled with the FW–H analogy, was specifically mentioned as being CPU-cost demanding.

5.2. Viscous CFD methods

As previously discussed, when potential methods are utilized, incorporating or modeling viscosity becomes essential before performing acoustic analyses. This is generally accomplished through semi-empirical models, which, compared to viscous computational fluid dynamics (CFD) solvers, often show greater uncertainties. As a result, many researchers employ viscous CFD techniques for cavitation and hydroacoustics studies, primarily using large-scale simulations such as Detached Eddy Simulation (DES), Improved Delayed Detached Eddy Simulations (I-DDES), or LES, and to a lesser extent, RANS simulations. The distinction between these methods lies in the extent to which turbulence is resolved. RANS/U-RANS turbulence modeling employs a mean-value approach to solve the time-averaged Navier–Stokes equations (NSE). From these fundamental equations, the RANS/U-RANS approach is derived by decomposing each solution variable into a mean and a fluctuating component, known as Reynolds decomposition. It is important to note that while these equations are typically further simplified by assuming incompressible flow in propulsion numerical problems, for hydro-acoustics modeling, the fluid must be considered weakly compressible. In contrast to the RANS approach, LES does not eliminate fluctuations by time-averaging the NSE. Instead, this method removes small turbulence scales using a spatial filter, solving the remaining scales. To filter out small scales from instantaneous solution

variables, a similar technique to RANS, called LES or filter decomposition (Deardorff, 1970), is employed. The key difference between RANS and LES filtering is that LES uses spatial filtering rather than time-average filtering. The unresolved or subgrid scales are typically modeled using various subgrid scale models, with the most commonly used being:

- **Smagorinsky subgrid-scale model**: This is regarded as the initial subgrid model created for LES simulations, serving as the foundation for various improvements. The primary limitation of this model is its high sensitivity to the model coefficient, which varies with local flow conditions. Furthermore, the turbulent viscosity does not disappear in laminar regions and requires adjustments near walls to achieve the correct viscous sublayer behavior. This issue is more pronounced in wall-bounded domains and complex geometries, directly impacting the computational time.
- **Dynamic Smagorinsky subgrid-scale model**: This model is an extension of the original Smagorinsky subgrid model. However, the coefficient in this model varies with time, thereby eliminating the need for damping functions.
- **WALE subgrid-scale model**: This is regarded as a contemporary subgrid-scale model. In this scenario, the WALE subgrid model coefficient does not have a universal value. However, the model is less sensitive to this coefficient compared to the Smagorinsky and Dynamic Smagorinsky models. Additionally, the WALE model does not require wall-damping functions to improve its accuracy.

A more computationally efficient alternative to LES is Detached Eddy Simulations (DES), which employ a hybrid RANS-LES method and were initially introduced by Spalart (1997). DES simulations are designed such that regions with non-rotational flows and boundary layers are addressed with a RANS method, whereas in regions of detached flow, a LES method is utilized. The switching criterion typically relies on a model coefficient (C_{DES}) that determines the turbulent length scale. This model coefficient generally has a fixed value and is derived through calibration based on the decay of homogeneous and isotropic turbulence (Spalart et al., 2006).

To prevent the resolved structures from penetrating the boundary layer/irrotational flow regions, a shielding function has been employed in this approach. Consequently, Spalart et al. (2006) introduced a modern formulation known as Delayed Detached Eddy Simulations (DDES). Subsequently, Shur et al. (2008) developed the latest RANS-LES hybrid formulation, termed Improved Delayed Detached Eddy Simulations (I-DDES).

These three types of turbulence approaches have been widely used for hydro-acoustics analysis of marine propellers.

5.2.1. RANS turbulence modeling

Focusing on research using RANS methodology, numerous studies stand out. Bertetta et al. (2012b) conducted a hydroacoustic optimization of the CPP propeller using genetic algorithms, considering variables such as chord, pitch, maximum camber, and maximum thickness. This study emphasized the importance of pitch and maximum camber distribution in cavity extension. The need to cover a wide range of propeller speeds to optimize noise production was also highlighted. They investigated frequencies up to 10^4 Hz and validated the noise results against previous predictions by BEM and cavitation tunnel experiments. Their validation indicated that the cavitation patterns were comparable, except for TVC slip-streams, which are difficult to capture with non-dynamic meshing techniques. K_T and K_Q were consistent, and in terms of SPL, although the agreement with the experiments was moderate, significant background noise levels were observed above 1000 Hz. Regarding the numerical efficiency of their framework, a computational cost of 3 h was reported for a converged solution.

In contrast to Bertetta et al. (2012a), the RANS approach was not yet considered computationally prohibitive for optimization, with a consumption rate of 20 h using 12 CPU processors for a converged solution. It is important to note that this latter study included a detailed verification with four grid solutions for the flow field solution, and the cavitation patterns were validated. However, a mesh sensitivity analysis for cavitation was still not reported. In a subsequent study, Lloyd et al. (2015b) compared the results obtained by the RANS approach with those from the FW–H acoustic analogy, showing that both methods were comparable in predicting blade pressure pulses. Notably, this contribution reported numerical uncertainties for both the flow field and the acoustic field solutions. Although the numerical uncertainties for the flow field were below 2.5%, those for the pressure probes exceeded 20%. The time consumption rates for the developed framework were not reported in this case.

Ghassemi et al. (2018) utilized a viscous CFD solver for hydrodynamic analysis, integrating it with an acoustic analogy that used the CFD output as input. They identified various noise sources in propellers and examined the impact of tip rake angle, which is particularly important in unconventional geometries, on sound pressure levels. By employing a RANS solver combined with the FW–H acoustic analogy porous formulation, they found that increasing the rake angle can decrease both propeller efficiency and sound pressure levels. Although their insights into URN generation mechanisms and mitigation are valuable, further research is required on the numerical efficiency of the proposed framework. Similar to several other studies, they did not report the assessment of numerical uncertainties and the time consumption of their method.

More recently, Jeong et al. (2022) used a RANS solver to study the effects of skew and rake angle on cavitation in submarine propellers, discovering that these angles delayed cavitation inception speed by more than 2 knots and reduced acoustic pressure. Their framework was validated against the experiments, showing a maximum deviation of 0.90% for the flow field solution (K_T and K_Q), indicating the accuracy of their framework. Various grid resolutions were employed, demonstrating satisfactory convergence of the flow-field solution, although numerical uncertainties and time consumption rates were not reported. As in other reviewed works, the convergence of the acoustic-field solution was not analyzed in detail and separately from the flow-field solution. Notably, TVC and TVC noise were analyzed and extracted based on vorticity (Q-Criterion) instead of the vapor volume fraction, accurately capturing the TVC streams. From a design perspective, the conclusions of Lidtke et al. (2016a) are significant. They emphasized the potential of RANS and U-RANS simulations coupled with the FW–H analogy for initial design stages because of their time efficiency, while highlighting the higher accuracy of scale-resolving simulations for detailed designs.

5.2.2. DES turbulence modeling

When analyzing cavitation and the noise it generates, it is crucial to recognize that the vapor bubbles created due to cavitation are subgrid scales (Hickel, 2015). Since cavitating noise is the main source of URN in marine propellers, the most accurate method for hydrodynamic analysis is DES/LES. The difference between DES and LES methods lies in the application of URANS equations in near-wall regions in DES to minimize computational costs.

In the field of Detached Eddy Simulations (DES), numerous advancements have been made recently. Shin and Andersen (2015) analyzed three different tip loading propellers, noting that reduced tip loading can lead to cloud cavitation, which in turn generates shock waves, noise, and blade erosion. They employed a DES approach along with a Rayleigh-Plesset cavitation model and concluded that increasing tip loading can reduce cloud cavitation. Regarding the numerical efficiency of their framework, a thorough evaluation of numerical uncertainties and mesh sensitivity was not provided. Subsequently, Hu et al. (2021a) utilized a DES approach combined with an FW–H analogy equation to investigate cavitating noise, emphasizing the significance of the receiver's location and the directivity graphs in Contra Rotating

Propellers (CRP). They used a Schnerr-Sauer cavitation model, which is based on the Rayleigh-Plesset model, and assumed the acoustic quadrupole term to be negligible. In their study, a series of receiving points was used and frequencies up to 10^4 Hz were examined. Compared to the experimental data, deviations of 0.52% for K_T and 9.75% for K_Q were observed. Their analysis revealed that the first five blade harmonics, below Hz, were the most energetic, and cavitating noise showed lower sensitivity to the receiver's location. Concerning the numerical efficiency of their framework, three different grids were used, demonstrating a satisfactory progression of mesh convergence. However, detailed reports on numerical uncertainties and time consumption rates were not provided.

In subsequent studies, Hu et al. (2021b) investigated the effect of skew on Tip Vortex Cavitation, which is the primary source of cavitating noise, utilizing the LES approach with a dynamic Smagorinsky subgrid model and the same cavitation model was used. Their research employed a dynamic meshing technique based on vorticity (Q-Criterion) rather than the volume fraction of vapor to capture TVC, demonstrating high effectiveness. In particular, this work provided a comprehensive account of the numerical uncertainties for the K_T, K_Q, and the vapor volume fraction. The uncertainty ranges were 0.39–1.06% and 0.28–0.79% for K_T and K_Q, respectively, and 4.24–13.07% for the vapor volume fraction. Although time consumption rates were not explicitly mentioned, the developed framework is known to be computationally intensive, typically necessitating HPC resources. Comparison errors were reported solely for the flow-field solution.

Recent studies have further highlighted the importance of a comprehensive analysis of TVC in resolving the URN problem. Belhenniche et al. (2024) investigated the impact of pores in propeller blades on mitigating TVC. They employed an E779-A model for their purely numerical study, integrating viscous CFD with FW–H analogy and the Schnerr-Sauer cavitation model. Their findings indicated a 10 dB reduction in SPL when comparing smooth and porous propellers, attributed to the suppression of tip vortices. However, this reduction in noise emissions came with efficiency losses ranging from 2.6% to 4.05%. As with other studies reviewed, the assessment of numerical uncertainties was not detailed. The time consumption rates were also not reported, but the use of a non-dynamic meshing technique with helicoidal refinements for the TVC regions suggests a high time cost. Cheng et al. (2024) utilized a similar numerical framework, but implemented an LES turbulence approach with a Dynamic Smagorinsky subgrid model. They examined a Postdam propeller model under both non-cavitating and cavitating (TVC) conditions, revealing a 20 dB difference between the two conditions across all frequencies studied.

It has been noted that the DES method can sometimes face problems resulting in artificial separation (as mentioned by Gritskevich et al. (2012)). Because of this issue, many researchers prefer using the Delayed Detached Eddy Simulation (DDES) or Improved Delayed Detached Eddy Simulation (I-DDES) methods, as introduced by Spalart et al. (2006) and Shur et al. (2008). These methods were employed by Stark and Shi (2021) for a hydroacoustic study of bioinspired tubercles (a sinusoidal waveform tubercle placed at the duct's inlet) in ducted propellers combined with an FW–H analogy formulation. This device was found to reduce the acoustic signature of ducted propellers by up to 11 dB. In this study, the Farassat porous formulation, where quadrupoles are modeled as pseudo-monopole and pseudo-dipole sources, is used, focusing on SPL and directivity as crucial parameters for the acoustic analysis.

Lidtke et al. (2022) focused on predicting noise, employing an I-DDES approach coupled with a mass transfer cavitation model (Schnerr Sauer) for this purpose. Their results, compared with experiments conducted in a depressurized wave basin, showed good agreement, though they suggested further research is needed for far-field noise prediction methods. The comparison of cavitation and noise predictions with experiments is noteworthy. In their study, the framework implemented evaluated TVC using a vorticity-based criterion (Q-Criterion) instead of vapor formation, yielding satisfactory comparisons of cavitation patterns. For the acoustic analysis, the frequency range extended up to 10^4 Hz, and background noise measurements were presented. While a detailed analysis of discretization, parameter, and iterative uncertainties was not provided, the statistical uncertainties of noise were estimated, with bounds of 5 and −3 dB. Similar to other examples, the time consumption rates for this framework were not reported.

In a more recent study, Gaggero and Ferrando (2023) explored wake instabilities in a CLT propeller using the I-DDES approach. Subsequently, the Boundary Element Method (BEM) techniques were utilized to optimize the blade geometry concerning pressure distribution. The study highlighted the importance of high-fidelity Computational Fluid Dynamics (CFD) methods when optimizing parameters beyond propulsive efficiency. Consistent with other reviewed works, cavitation was assessed based on vorticity (Q-Criterion). A significant insight from this work was the comparison between standard CLT and new-generation CLT geometries, which revealed that secondary vortices in standard CLT geometries were reduced when the tip bending was smoothed.

Finally, in terms of the efficiency of the implemented numerical method, a comprehensive evaluation of the numerical uncertainties and the time cost was not provided. In addition, the contributions of Sezen and Atlar (2023a,b,c) should be highlighted. Their research utilized a DES approach with a Schnerr-Sauer cavitation model combined with the FW–H analogy to evaluate hydroacoustic performance under both cavitating and non-cavitating conditions and various inflow regimes. In their investigations, the frequency range extended up to 10^4 Hz. Upon validation of their results, a satisfactory agreement was noted, with some deviations of 5–10 dB observed at lower frequencies in model scale. In full scale, the deviations increased significantly, reaching differences of 100 dB between sea trials and simulation results. It was recommended to use relatively small permeable surfaces when employing a porous formulation to improve noise capture accuracy. From these contributions, it is noteworthy that they implemented a vorticity-based dynamic meshing technique, which effectively captured TVC. Regarding numerical discretization, although a detailed assessment of mesh sensitivity and time costs was not provided, the critical importance of performing verification in hydro-acoustic simulations is clearly emphasized. More recently, Köksal et al. (2024) introduced a dynamic meshing framework specifically targeting TVC noise. Their study involved a 4-level adaptive mesh refinement based on the vapor fraction and the Q-Criterion. When applied to a Kings College D Series (KCD) propeller model, it demonstrated a reasonable comparison of the URN spectrum, particularly in capturing the TVC hump at high frequencies (>1 kHz). The impact of TVC noise was detected from 100 Hz onwards.

The findings reported by Chen et al. (2023) are noteworthy. Their research revealed that the initial 10–12 mode shapes of the propeller contained 99.8% of the energy, with the first 5 modes significantly contributing to the noise spectrum and tonal peaks. They also highlighted the influence of complex flow conditions on the growth of URN.

5.2.3. LES turbulence modeling

With regard to LES methodologies, numerous researchers have utilized them for hydroacoustic studies and the optimization of propellers. In this context, Lidtke et al. (2016b) introduced a precise method for identifying primary noise sources and characterizing sheet cavitation, which involved a LES approach with the Schnerr-Sauer cavitation model, integrated with a porous formulation of the FW–H acoustic analogy. From a theoretical perspective, the work by Cianferra et al. (2019) is highly enlightening, as it outlines the most commonly used subgrid-scale models, namely Smagorinsky and dynamic Smagorinsky, and emphasizes the necessity of a porous formulation when applying the FW–H analogy in hydroacoustics. The importance of subgrid-scale modeling in LES approaches (as in DES/I-DDES) is crucial, as demonstrated by Zhang et al. (2010), who employed a Smagorinsky subgrid-scale modeling approach in their study of cavity-induced noise. In their

developed framework, it should be noted that for URN predictions, they achieved comparison uncertainties with experimental results of 7%–10% in frequency and magnitude in SPL. It should be noted that the test case geometry in the mentioned study was not complex, potentially increasing the computational cost of implementing a standard Smagorinsky subgrid-scale model. Additionally, it is important to highlight that, as stated in several outlined works, a detailed assessment of numerical uncertainties, mesh sensitivity, and CPU cost was not reported.

Other subgrid-scale models have also demonstrated their accuracy, such as the WALE subgrid model, which was used by He et al. (2020) to numerically predict hydrofoil noise. For the acoustic analysis, the FW–H analogy was employed, using Kirchhoff integrals to model quadrupole terms. The acoustic analysis performed is noteworthy, as they not only analyzed the SPL spectrum up to the kHz order of magnitude (5000 Hz) but also conducted a 1/3 Octave band analysis, revealing how acoustic energy is distributed across frequencies, with a clear dominance of higher frequencies (typically due to cavitation nonlinearities) in the acoustic power. Furthermore, they provided valuable information on the effects of the inflow angle of attack on cavitation and noise. Regarding the numerical scheme implemented, their manuscript did not present a detailed study of convergence and CPU cost was not presented in their manuscript. Balaras et al. (2015) utilized a Large Eddy Simulation (LES) approach to investigate submarine propellers. However, it was acknowledged that this approach could be computationally intensive. To save computational resources, a grid relaxation method was introduced along with the Immersed Boundary method for solution reconstruction. Their approach resulted in deviations of approximately 2% in open water simulations, with reported similarities in the vorticity patterns observed in experimental results.

Bensow and Liefvendahl (2016) investigated the hull–propeller configuration of a twin screw ferry using an LES method coupled with a FW–H analogy porous formulation. The developed setup should be highlighted as they managed to integrate viscous CFD for self-propulsion simulations under cavitating conditions, which is already notably time-consuming, with FW–H acoustic analogy. Although specific time consumption rates were not provided, the time step required to achieve a Courant number around 1 was $\Delta t = 5.3 \cdot 10^{-6}$, which is an order of magnitude smaller than in simulations involving only the propeller geometry. For the acoustic analysis, they covered a frequency range up to 10 kHz, with the highest deviations (around 20 dB) occurring at frequencies below 500 Hz. Regarding numerical discretization, the verification process was not detailed in their manuscript. More recently, Kimmerl et al. (2021) performed an analysis using the Large Eddy Simulation (LES) method with the standard Smagorinsky subgrid model, combined with the Ffowcs Williams–Hawkings (FW–H) analogy for noise prediction. The Kirchhoff integral was included to account for quadrupole terms, and the Schnerr-Sauer cavitation model was employed. Their findings showed strong agreement in evaluating near-field noise, with a conservative SPL estimation observed at low frequencies (below 1000 Hz). It is important to note that there was a lack of validation for far-field noise predictions. Regarding their numerical setup, it is notably innovative that they used four different types of grids for mesh generation, reaching up to 30 million cells. Numerical uncertainties and time consumption rates were not specified in the manuscript.

In addition, it is important to highlight the significant contributions made in the studies by Posa et al. (2022a,b,c,d) and Posa (2022). These studies produced results that reasonably matched the experimental data, with deviations ranging from 10–20 dB for both low and high frequencies. A large eddy simulation (LES) method, using the WALE subgrid-scale model, was employed, combined with the Ffowcs Williams–Hawkings (FW–H) analogy porous formulation. These studies thoroughly examined the acoustic properties of propellers and their interactions with upstream hydrofoils, reflecting real-world scenarios involving propeller–rudder structures. In terms of hydro-acoustic modeling, it should be noted that they covered frequencies up to 100 kHz, which is the upper limit recommended by class rules for vessel noise characterization. With regard to numerical discretization, they provided valuable information on the number of elements and time consumption rates. Specifically, their grid resolution reached 840 million points, requiring HPC resources (1.5 million hours at 1024 cores) for a converged solution. A detailed verification process was not reported in their publication.

Lastly, it is crucial to underscore the contributions of Dlamini et al. (2023). An LES method, with the standard Smagorinsky scale model integrated with the Williams–Hawkings (FW–H) acoustic analogy, was used alongside the cavitation models. It is worth mentioning that they covered a frequency range from 10 Hz to 80 kHz, aligning with the frequency recommendations for vessel acoustic characterization from various class notes. However, there was a significant limitation in mesh sizes (they reached only up to 1 million cells), which resulted in slightly higher deviations compared to experimental results from other published research, up to 20–30 dB.

Finally, the methodology used for acoustic modeling requires further exploration. Unlike the various approaches used for flow analysis, there is a broad consensus in favor of the acoustic analogy. This method computes noise sources within a specified region and subsequently employs a wave equation to determine the noise propagation. As such, it effectively separates sound generation from sound propagation, as emphasized by Lloyd et al. (2014). In this regard, as previously mentioned, the most prevalent acoustic analogy is that formulated by Ffowcs Williams and Hawkings (1969). This approach classifies sound generation sources into three linear terms: thickness and loading noise, along with two non-linear terms, as described by Lloyd et al. (2015a).

In the domain of aeroacoustics, primarily the first field of application for acoustic analogies, and within the context of low Mach numbers and subsonic regimes, the non-linear terms are often omitted. Nevertheless, as Ianniello et al. (2013) have explained, in aeroacoustics, non-linear phenomena and shock waves are not expected to arise until the Mach number surpasses 0.9. In contrast, in hydroacoustics, unsteady cavitation and shock waves may occur even at low propeller rotational speeds, necessitating the consideration of quadrupole or non-linear terms. On the other hand, considering the non-linear terms directly can increase the computational expense considerably, which justifies the use of porous formulations (Farassat, 1975; Brentner, 1986; Di Francescantonio, 1997), which assume that the non-penetration requirement for FW–H is not required and take the quadrupole terms into account in an implicit way.

Furthermore, the study presented in Soydan and Bal (2021) is extremely pertinent to the current discussion, offering insights beyond the specific sub-topics addressed in the cited numerical studies. In their research, they examined scale effects under both cavitating and non-cavitating conditions, employing a lifting surface (potential) method for the latter and a viscous CFD approach for the former. They investigated a DTMB-4119 propeller model with varying scale factors from 1 to 6. They developed a semi-empirical model using logarithmic regressions for propellers with $\lambda \leq 2$. The numerical results for thrust and torque for propellers with $\lambda \geq 2$ were compared with ITTC models and their semi-empirical models, revealing that the regressions offered better predictions than ITTC. Although further research is required, particularly concerning extrapolation methods for URN levels, this study is highly relevant and aligns with the statements in ITTC (2021b) about the significance of creating accurate full-scale extrapolation models.

Considering the studies reviewed, several remarks can be made to highlight areas where additional research could be advantageous. Previous and ongoing investigations focus mainly on the integration of acoustic analogies with flow solvers. Despite the interest in advancing numerical frameworks for hydroacoustic analysis, there is a notable

lack of comprehensive studies on the numerical uncertainties and mesh sensitivity of acoustic analogies. Recent studies (Portillo-Juan et al., 2024b,a) have underscored the importance of numerical uncertainties in hydroacoustic analyses, particularly for the implementation of these frameworks in design applications. Furthermore, it has been noted that research specifically targeting the numerical prediction of cavitating noise from TVC is relatively novel, especially when incorporating dynamic mesh techniques with acoustic models.

6. Optimization methods

As already introduced, the implementation of optimization techniques for an integrated design of propellers is growing vastly. The present section aims to review the evolution of the different optimization frameworks that have been used for the design of efficient and silent propellers.

6.1. Iterative optimization

For a long time, propeller design optimization has been practiced, focusing primarily on efficiency. However, multi-objective optimization that includes noise for marine propellers has rarely been addressed. Nonetheless, certain studies on cavitation improvement via iterative optimization processes are noteworthy. Kuiper and Jessup (1993) serves as a notable example. Their research involved iterative optimization of the design of the propeller blade section, targeting TVC inception speed as the objective function. This resulted in a delay in Tip Vortex inception and a slight increase in K_T and η_0. Kerwin (1994) conducted a similar study using the panel method, which allowed the analysis of only sheet cavitation. Building on the concept of the impact of profile design on propeller cavitation, Dang (2004) employed an iterative procedure between 2D profile design and 3D propeller code to optimize the cavitation bucket and reduce the load on the leading edge.

Bal (2011) conducted a comparable optimization process, utilizing a lifting surface method for solving the flow field and targeting the radial distribution of circulation, η_{MAX}, and T as objectives, resulting in significant reductions in circulation with similar propeller efficiencies. However, there remains an interest in straightforward optimization methods such as low-fidelity or semi-empirical models, particularly during the conceptual design phase. For example, Lafeber et al. (2022) developed a low-fidelity model for URN estimation based on 24 000 cases. In their study, pressure forces were calculated using a BEM solver, and semi-empirical models were employed to estimate noise. This information was used to construct a low-fidelity model to obtain a preliminary noise spectrum during the concept design stage. The research conducted by Kinnas et al. (1998) and Griffin and Kinnas (1998) serves as a strong precedent for optimization studies that consider both propeller efficiency and cavitation. These studies implemented a non-linear optimization framework (CAVOPT-3D) where efficiency was the objective function, and cavitation volume and tip skew were constraints. To solve the flow field, a VLM-based method was used. By employing this framework, which is theoretically a single objective but includes cavity volume and surface as design constraints, a 25% reduction in cavity volume with a 1.2% η_0 increase was achieved for the case study. More recent studies have also been reported (Kinnas et al., 2021), using wake distribution as a constraint.

6.2. Genetic type optimization

As previously mentioned, Artificial Intelligence (AI) has significantly improved multi-objective optimization processes in propeller design, particularly in optimizing cavitation and noise generation while maintaining efficiency. One widely used method for multi-objective optimization with AI is the application of Genetic Algorithms. These algorithms are inspired by natural evolution, starting with an initial

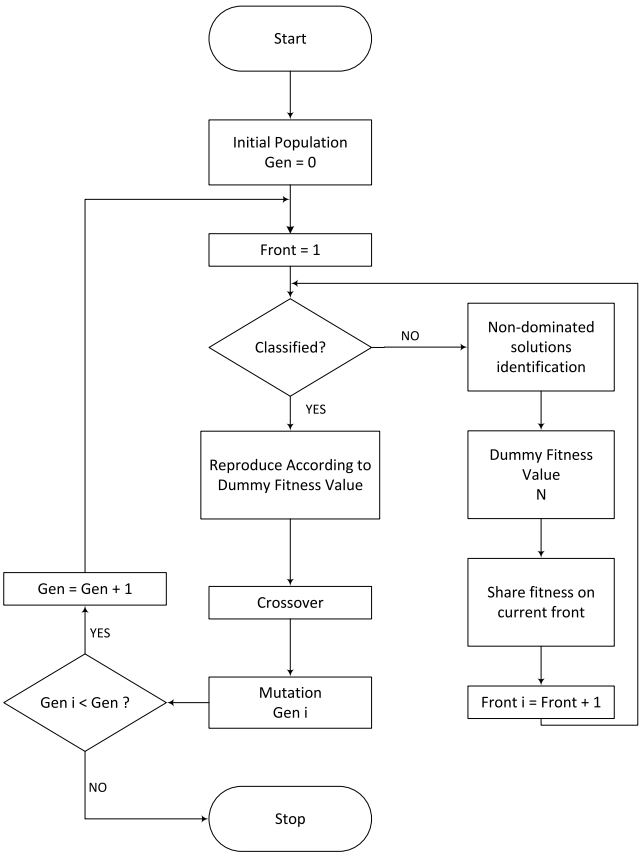


Fig. 6. NSGA framework.

population of potential solutions. After each iteration, the best candidates from the mutated population are chosen to form a new set until an optimal solution is found. Unlike single-optimization problems, where a single 'best' solution is selected, Genetic Algorithms retain a set of potential solutions within the Pareto front, forming a population. There are various algorithms for multiobjective optimization problems, but for maritime applications, it is crucial to consider that the optimized function may have multiple peaks. Therefore, a commonly used approach is the Non-dominated Sorting Genetic Algorithm (NSGA) (Srinivas and Deb, 1994). This algorithm begins with a random population, which is assigned a dummy fitness value based on the population size. The initial front is initially disregarded and a second front is given a new dummy fitness value, which is lower. With these two populations, the processes of sharing, crossover, and mutation can begin (Bagchi, 1999). The general structure of NSGA is illustrated in Fig. 6.

Since this genetic algorithm was initially computationally intensive, subsequent advances have addressed this issue (NSGA-II by Deb et al. (2002) and NSGA-III by Jain and Deb (2013)). In the context of hydro-acoustic optimization of marine propellers, the works of Gaggero et al. (2015, 2016a) and Gaggero and Ferrando (2023) are particularly noteworthy, as they encompass the design, optimization, and analysis of an unconventional geometry. These studies clearly outline the crucial aspects to consider in an optimization process using genetic algorithms.

Firstly, it is essential to properly vary the free parameters and the objective functions. Since this review focuses on hydro-acoustic optimization, the primary objective functions will be propeller efficiency (commonly denoted as η_0) and SPL. Regarding the free parameters, a typical method is to consider geometrical parameters that influence both efficiency and noise generation. Once this classification is established, a common technique is to adjust the radial distribution of the

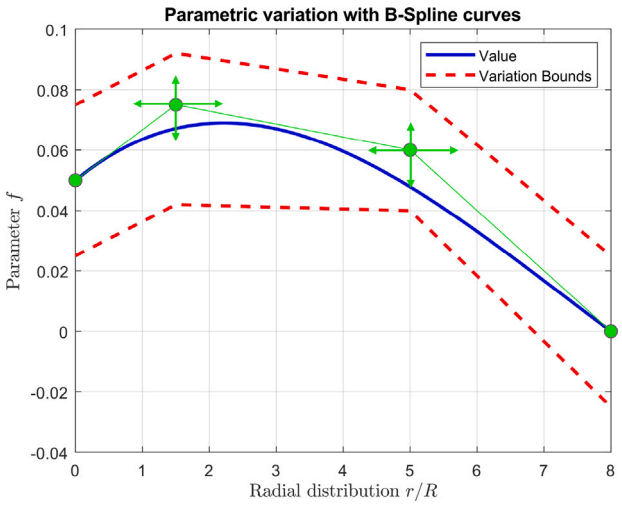


Fig. 7. Parametric variation range for camber distribution. Figure inspired from the optimization study performed with genetic algorithms at Gaggero et al. (2016b).

free parameters. To perform this task efficiently, these distributions are usually defined as B-Spline curves with n control points. For each curve, a variation range is set for $m \leq n$ control points (refer to Fig. 7). Subsequently, an initial population of geometries is generated by altering the original parametric description (control points) within the specified ranges.

The initial population is used to evaluate the objective function, leading to the optimized geometries. In Gaggero et al. (2016a), an initial population of 200 propellers was established, and after the optimization loop, a total of 10,000 geometries, or 50 generations, were examined. Doijode et al. (2022a) segmented the propeller design process into clusters. In their initial study, they parameterized the propeller blade using a B-Spline surface and generated the geometries with a Halton sampling sequence. They developed a dynamic optimization framework incorporating both supervised and unsupervised classifiers. Unsupervised classifiers analyze the design space to identify clusters of similar geometries, while supervised classifiers determine which clusters can reduce the number of evaluations, making them advantageous in the design space. In this initial study, the population was generated over a B4-70 propeller model with a P/D value of 1.0, achieving an optimum with a 2.56% η_0 increase. In the second phase of their research (Doijode et al., 2022b), the NSGA-III genetic algorithm was utilized, with Naive-Bayes classifiers reducing the evaluations by up to 30%. In this phase, they achieved an efficiency of 52.02%, representing a 10.84% improvement over the original propeller. In their research, a BEM solver was employed for hydrodynamic performance and cavitation analysis, and the optimized design showed no cavitation on the pressure side, reduced suction side cavitation, and acceptable tip vortex cavitation.

Gypa et al. (2022) highlighted that optimizing blades using genetic algorithms can become arduous when cavitation also needs to be optimized, as designer evaluation might be necessary. As a result, due to the large populations generated in these design loops (sometimes exceeding thousands of models), such scenarios can lead to designer fatigue. To minimize the number of evaluations, they integrated an ML pipeline with the optimization framework, reducing designer evaluations to data labeling, thereby only eliminating unfeasible options. In their study, they compared three different ML algorithms and applied their framework to five different propellers. After this process, to assess the accuracy of the different algorithms, they implemented Nested Cross Validation (NCV), as presented in Stone (1974), with two loops. In the outer loop, the dataset generated in the framework is divided into training and testing sets. The training sets are used as input for the inner loop, while the testing sets are reserved for validation. The inner loop determines the best hyperparameters for the ML algorithm's accuracy. The entire procedure demonstrated an accuracy in predicting cavitation nuisance between 0.87 and 0.98. A notable prior research effort is that of Mulcahy et al. (2014), where a Sequential Least Square Programming (SQLSP) algorithm was used to optimize composite propeller noise, resulting in a 25% reduction in generated noise in the optimized geometry, (achieved through load shape-adaptive blades), compared to the original, and suggesting that acoustic optimization can be approached through geometrical changes.

Aytaç and Zafer (2024) present an innovative approach to hydroacoustic optimization. They integrated Generative Adversarial Networks (GAN) with noise optimization networks to enhance a NACA-0009 2D foil. Their method, termed Noise-GAN, was compared with Bezier-GAN optimization frameworks within a 0–15° angle of attack range. They demonstrated a 10 dB improvement in their framework over the Bezier-GAN frameworks. When comparing their optimized foil with the original NACA-0009 and the Bezier-GAN optimized version, improvements reached up to 60 dB at 5 kHz for the 3° AoA, along with the highest C_L/C_D values for all analyzed cases. However, it is important to note that 3-D effects were not considered, and the study was entirely numerical, thus experimental validation of the results was not possible.

6.3. Data-driven optimization

The methods discussed in this section reveal that the direct application of genetic algorithms to hydroacoustics significantly increases computational costs. The BEM approach is used more commonly for hydrodynamic studies but is less accurate than scale-resolving simulations, particularly since cavitating noise is the primary noise source in marine propellers. Jiang et al. (2020) highlighted the issue of high computational costs in large-scale simulations using genetic algorithms and explored the potential of the Data Driven Evolutionary Optimization Method (DDEOM) to mitigate noise from marine propellers. This method acknowledges that defining the objective function can be complex for some optimization problems and uses data from previous simulations and experiments to create surrogate models (Jin et al., 2018). Jiang et al. (2020) employed a neural network trained with a simulation-based surrogate model and compared its performance with a traditional evolutionary optimisation method (TEOM), finding that the time cost of DDEOM was only 0. 68% of TEOM. The key aspect of data-driven evolutionary optimization in marine propulsor hydroacoustics is the development of a surrogate model that can significantly reduce the computational time needed for numerical simulations when analyzing acoustic and flow fields. This requires generating high-fidelity data to construct the reduced-order model. A general framework for a data-driven optimization process is depicted in Fig. 8.

Artificial Neural Networks (ANN) are a particular kind of surrogate model that has been effectively applied to multi-objective optimization processes. According to Walczak and Cerpa (2003), ANN emulate the electrical activity of the nervous system. Therefore, input values are linked to processing units called neurodes. These neurodes can be further connected to other neurodes, ultimately producing the output values. The neurodes are organized into layers, referred to as hidden layers. The connections between units are represented as weighted connections. Finally, it is essential to specify the learning method, which may be supervised or unsupervised. In the case of the former, the output value for ANN training is derived during the training process itself. Conversely, for the latter, the output value for training is provided. A general ANN structure is illustrated in Fig. 9.

Šegota et al. (2020) utilized an Artificial Neural Network to predict the torque of a twin screw frigate propeller. In their study, they trained the ANN with 7200 starboard and 7237 port propellers, respectively, achieving a Coefficient of Determination (R^2) value of 0.99 for the torque estimation regressions. The potential of Data-Driven Methods (DDM) was also explored by Miglianti et al. (2019); however, in this

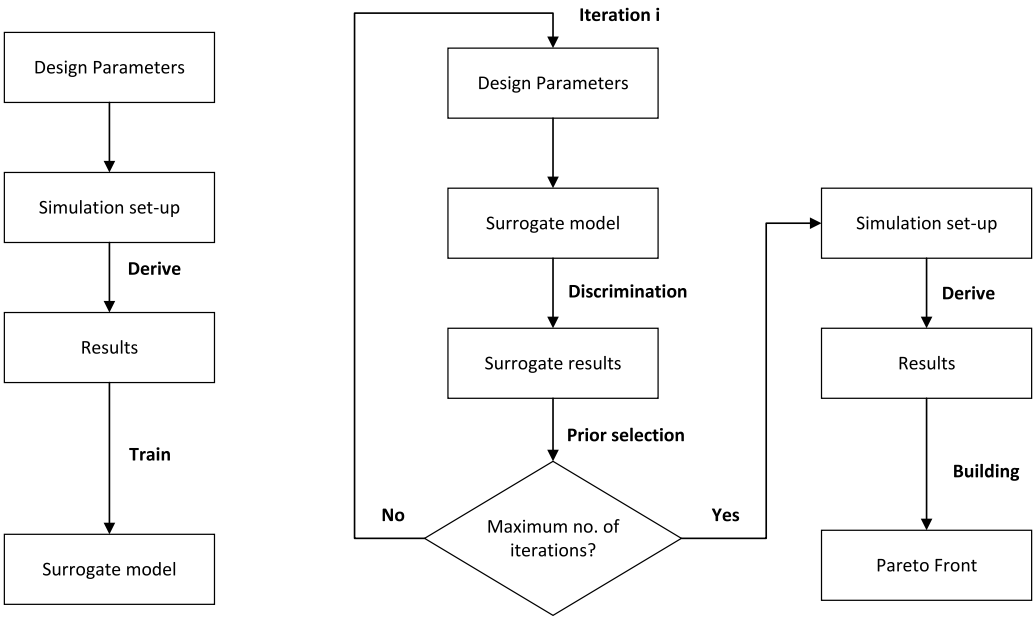


Fig. 8. Data driven optimization general framework. Inspired by Jiang et al. (2020).

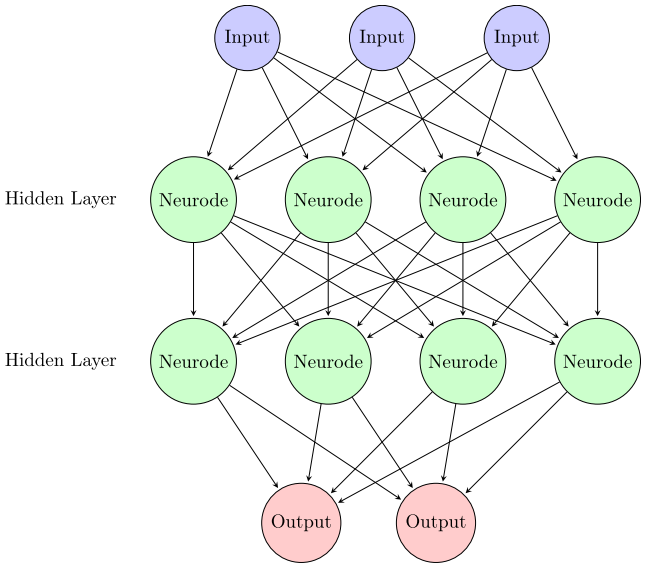


Fig. 9. Artificial neural network general framework.

initial study, DDM was applied solely to broadband noise, neglecting tonal peaks. Building on the demonstrated potential, they later employed a deep learning approach that included both broadband and tonal noise in subsequent works (Miglianti et al., 2020; Oneto et al., 2020). These studies involved three controllable pitch propellers and predicted the cavitating noise using DDM constructed with BEM simulations and semi-empirical models. Another notable contribution is by Kou et al. (2023), who used auto-encoders (ANN) to optimize the aeroacoustics of an airfoil shape. Additionally, the work by Kimmerl and Abdel-Maksoud (2023) on URN numerical analysis assessment is noteworthy; besides employing FW–H acoustic analogy, they also presented various noise evaluation methods beyond SPL spectra, including directivity maps, Q-Criterion isosurfaces, and pressure pulse graphs. More recently, Li et al. (2024) employed convolutional neural network, to predict the velocity and vorticity fields generated by cylinders. Their model simplified the fluid dynamics, showing good comparison with the CFD predictions, with a coefficient of determination of 0.96 to >0.99 for all the cases studied. In a similar study, Xie et al. (2024) used an auto-encoder (built with CFD data) to predict the wake-field of a submarine. In their results, their neural network showed a mean square error of 0.00168 when compared with CFD predictions with a coefficient of determination of $R > 0.99$. The time consumption reached with their model is also noteworthy, as only 0.259 s (using a NVIDIA TITAN RTX GPU) was used for a test-case prediction. Despite the different approaches and the potential of Deep Learning over Genetic Algorithms, these models are typically constructed with BEM simulations combined with semi-empirical methods. Therefore, the feasibility of developing Deep Learning models with Scale Resolving Simulations warrants further investigation.

As discussed in the previous sections, it is advantageous to emphasize the less explored areas identified in the reviewed literature. From the various studies examined, it is evident that when AI is applied to multi-objective optimization processes, the predominant method has been the use of Genetic Algorithms, followed by Artificial Neural Networks (ANN). However, studies on surrogate-assisted optimization are limited, particularly when both energy and acoustic efficiency are considered as objective functions. In this context, and based on the observations made in the earlier sections, it can be noted that developing multi-fidelity models for designing more efficient and quieter propellers remains a relatively new approach. Furthermore, the studies that employ surrogate-assisted optimization typically construct their reduced-order models using data from conventional geometries. Unconventional tip-raked propellers (e.g., CLT, Kappel) exhibit different behaviors compared to conventional geometries in hydro-acoustics, often showing higher efficiencies with weaker cavitation patterns and/or reduced sound generation. Therefore, it can be inferred that tip bending introduces new geometrical parameters not usually considered in conventional geometries, as well as different physical behaviors. Thus, the authors deem it particularly important to develop reduced-order models specifically based on data from unconventional propellers to further enhance the hydro-acoustic performance of marine propellers within reasonable time constraints.

7. Conclusions

This paper provides an extensive overview of design strategies for hydroacoustic optimization of propellers. The research aimed to investigate the latest advancements in numerical and experimental methods

for propeller design, with an emphasis on combining hydrodynamics and acoustics to create efficient and silent propellers. The overview emphasized different design techniques and optimization strategies used in the domain, such as numerical simulations, experimental evaluations, and multiobjective optimization approaches. These methods help researchers and engineers balance the competing goals of propeller efficiency and noise reduction.

Based on the analysis of the reviewed literature, several important findings have surfaced. Firstly, there has been a noticeable increase in awareness of the threat of acoustic pollution, as evidenced by guidelines and recommendations from public authorities. Furthermore, these communications have taken on a more urgent tone over the past eight years, as seen in the latest IMO Guidelines. This trend may set the stage for future restrictive regulations in the medium-short term. Given that the primary source of noise from ships is the propeller, it is essential for the maritime industry to develop hydroacoustic optimization methods, particularly to meet future restrictions on acoustic emissions. Secondly, it is clear that optimizing the hydroacoustic performance of propellers requires a multidisciplinary approach, taking into account both hydrodynamic factors and radiated noise characteristics.

To design and optimize marine propellers with a focus on acoustic and propulsive efficiency, three primary tools have been identified: experiments, numerical simulations, and AI techniques. From the experimental group, two key elements have been recognized as milestones in conducting experimental campaigns to predict the Underwater Radiated Noise (URN). Firstly, the acoustic characterization of the facility is performed through defining the tunnel transfer function. Transfer functions describe the response of a specific cavitation tunnel to acoustic signals, thereby allowing the determination of the net noise produced by the propeller. Secondly, since noise generation is a highly stochastic phenomenon, it is essential to increase the number of measurements in these experimental campaigns. In large facilities, this has been accomplished by employing hydrophone planar arrays, which are also recommended by the ITTC. Conversely, some experimental facilities face high-dimension constraints, making the use of planar arrays impractical. Therefore, a viable alternative is to use a discrete number of arrays positioned on one of the cavitation tunnel test section walls, mimicking the concentric arrangement of planar arrays as closely as possible (e.g., three hydrophones arranged in a triangular shape or four in a rhombus shape). Additionally, to prevent reverberation, placing the hydrophones in acoustic chambers (constructed from materials with the same acoustic impedance as the tunnel windows) rather than inside the tunnel has proven to be advantageous.

In terms of numerical tools, the incorporation of sophisticated computational methods, such as computational fluid dynamics (CFD), has significantly enhanced the design process by offering precise prediction capabilities. In this context, the integration of CFD viscous solvers in large-scale simulations, which are inherently time-intensive, with acoustic analogies under cavitating conditions presents a major challenge due to the necessity of including cavitation studies in hydroacoustics numerical analyses. Concerning the application of acoustic analogies (originating from aeronautics) to the marine domain, specific considerations must be addressed. Firstly, the volume non-linear terms must be taken into account, as opposed to air, in marine scenarios, shock waves can emerge at very low Mach numbers due to cavitation. This has prompted the development of porous formulations that calculate all noise terms. Additionally, further investigation is required into the convergence and stability of numerical noise prediction.

Thirdly, the implementation of AI tools for the hydroacoustic optimization of marine propellers is highlighted as a crucial aspect in the design of non-traditional propeller geometries. Traditionally, genetic algorithms have been employed for the multi-objective optimization of propulsion devices, but they become impractical for large-scale simulations due to their high time consumption. An effective alternative that has demonstrated success in optimizing marine propellers in terms of both efficiency and acoustic emissions is the use of Neural Networks trained with surrogate models derived from simulation and experimental data.

Furthermore, several general observations and conclusions should be emphasized. Firstly, it has been noted that in terms of propeller blade design, there is a significant correlation between blade loading and noise production. This has shifted the focus towards unconventional geometries, particularly tip-raked propellers. The existing literature lacks a thorough examination of how altering the geometrical parameters of the tip bending (such as the bending angle) can achieve quiet and efficient propellers. Additionally, as previously mentioned, there is a lack of multi-fidelity frameworks for hydro-acoustic propeller optimization that integrate low-fidelity models, constructed with surrogate models, with high-fidelity models (such as scale-resolving simulations). The authors also find it crucial to highlight that, given the computational demands of high-fidelity hydro-acoustic simulations, it is essential to explore high-performance computing techniques to speed up and enhance the efficiency of these frameworks.

Moreover, considering the nature of the URN mitigation challenge and the existing measures, it is clear that a multidisciplinary approach is crucial to address this issue thoroughly. Indeed, the expertise gained from the aerospace industry in hydro-acoustics has shown the capability to quickly develop accurate noise prediction techniques. Drawing inspiration from this, the authors assert that incorporating knowledge from various disciplines can greatly improve the creation of effective silent propellers. Specifically, the authors emphasize the significance of insights from biology and oceanography in evaluating the effects of different technologies on marine life and pinpointing key frequencies for URN reduction. Furthermore, experts in mechanical engineering and particularly in sensing can make valuable contributions, especially in full-scale measurements.

By improving the comprehension and techniques of propeller design, this study helps create more efficient and quiet propulsion systems for various marine uses. The results shared in this document lay the foundation for future investigations, supporting ongoing efforts to improve propeller performance and minimize noise emissions along with their impact on marine environments.

CRediT authorship contribution statement

Adrian Portillo-Juan: Writing – review & editing, Writing – original draft, Resources, Methodology, Investigation, Formal analysis, Conceptualization. **Simone Saettone:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Investigation. **Poul Andersen:** Writing – review & editing, Supervision. **Esteban Ferrer:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Funding acquisition.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Adrian Portillo Juan reports financial support was provided by Polytechnic University of Madrid. Esteban Ferrer reports financial support was provided by European Union. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

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